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PROJECT

Comparison of Skyfield SGP4 and Basilisk Numerical Orbit Propagation for GNSS-R Formation-Flying Applications

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AI DECLARATION

ABSTRACT

SAMMENDRAG

PREFACE

Write the preface of your thesis here.

You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

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ABBREVIATIONS

Strongest narative for my thesis:

1. My specialization project showed that single-TLE SGP4 is not sufficient as ground truth, and that simplified drag limited the conclusions.
2. The Master's thesis improves upon the validation reference by using regularly update TLEs, making Skyfield-SGP4 a short-horizon, observation-updated reference
3. Basilisk is then validated as sufficiently accurate for long time-horizon mission simulation
4. The validated Basilisk propagator is used inside a realistic two-satellite GNSS-R formation simulator
5. Mission lifetime is estimated from fuel depletion + formation loss, with battery/power constraints shaping operations.

Scope: Focus on:

- Basilisk propagation validation
- Two-satellite deployment and formation acquisition
- Autonomous mode switching
- battery and fuel lifetime curves
- formation evolution
- mission lifetime estimate

Include, but keep on a high level:

- GNSS-R signal physics (just what is required to enable observation)
- Advanced controller derivations (both attitude and formation)

REMEMBER:

- Go over glosseries, and remove any that is only use once. Also make sure that 'gls' is used consistently for the same terms
- Use Inspira's 'plagiat' checker to verify that the theory sections ported from my specialization project report has been modified enough to be usable.
- In "Natural drift under Ideal two-body motion" I use the subscript " L " for the leader satellite, which has previously been denoted using " l ". I would just use the latter here too, but I cannot use " f " for the follower, since this denotes the RTN frame. Therefore: use L for leader, and change the use throughout the text, INCLUDING the figures.
- In "Natural drift under Ideal two-body motion", the vector for orbital elements is denoted using α , which hasn't been declared previously. Add this in the "Classic orbital elements" section.
- SET "attControlTime" PARAMETER FOR spacecraftReconfig MODULE
- I think I have been very inconsistent in the use of capitalization in regards to the different simulators. I should avoid capitalization completely.
- Write a small paragraph about how classic orbital elements become undefined at near-circular or -equatorial orbits
- verify process-task-model structure in [.0.3](#)
- If time: Create a table showing all the boolean "switching parameters" and their true-conditions definitions

INTRODUCTION

2.1 Background & Motivation

Global Navigation Satellite Systems (GNSS) have become deeply embedded into our modern infrastructure with multiple constellations, such as GPS, GLONASS, Galileo and BeiDou, ensuring that four satellites are observable from the surface at any given time, thus providing worldwide coverage [nasa_earthdata_gnss]. The sheer number of operational satellites that broadcast GNSS signals makes it one of the most powerful signals of opportunity for remote sensing, enabling monitoring without the need for onboard transmitters.

At the same time, the space sector has recently undergone rapid changes. Launch costs have fallen significantly due to government outsourcing of launch providers to private companies, and the competitive environment it produced. This is clearly shown in figure ??, where the per-kilogram launch cost has hit an all-time low with the deployment of SpaceX rockets *Falcon9* and *Falcon Heavy* [jones2018launch_cost]. As a result, it has become increasingly feasible for universities, research groups and smaller nations to deploy space-based systems dedicated to leveraging GNSS signals for remote sensing applications.

In parallel with this growth, there has also been an increase in the frequency of reported intentional GNSS interference to disrupt aviation operations. The interference can be divided into separate categories characterized by their difference in method and effect: *Jamming* refers to the relative high-power emission of Radio Frequency (RF) noise to block GNSS signals, while *spoofing* is the transmission of signals that resemble those from real satellites, misleading the GNSS receiver to think it has a different position [ffi_factsheet_2024]. The Norwegian Defense Research Establishment (FFI) states that: "In Norway's northernmost county of Finnmark, close to the Russian border, loss of GNSS signals due to jamming occurs nearly daily. [...] The number of disruptions has increased significantly since Russia's invasion of Ukraine in 2022." [ffi_jamming_2024]. Data from *gpsjam.org* suggest that similar disruptions have been reported across Eastern

Europe, the Middle East, and the Arctic [gpsjam_2025]. These events highlight a broader geopolitical trend: GNSS interference is becoming more common. This places increased strategic value on systems that can detect and localize jamming and spoofing events.

GNSS Reflectometry (GNSS-R) offers a promising technique for addressing these challenges. According to Jin and Komjathy [jin2010gnss-R], Hall and Cordey [hall_cordey_1988] were the first to propose that GNSS L-band reflections could be exploited in a bistatic radar configuration to retrieve ocean surface properties, laying the foundation for modern GNSS-R techniques. Recent theory suggest that the same bistatic configuration can be used for surveillance of the GNSS spectrum, where multiple GNSS-R satellites in formation can detect and localize sources of jamming and spoofing. This emerging use-case is one of the central research goals of NTNU SmallSat Lab’s GNSS-R project, which aim to develop methods for this process [ntnu_smallsat_gnssr_2025]. The project currently investigates this concept through its GNSS-R satellite mission study, exploring formations geometries, orbit configurations and system designs that would enable robust detection of GNSS interference sources and ships.

Accurate knowledge of spacecraft position and velocity is essential for the performance of any GNSS-R system. In such systems, regardless of application, the reflected GNSS signal is interpreted by comparing its time delay and Doppler shift relative to the direct signal. This process is highly sensitive to the specular reflection point’s geolocation, which is given by the geometric relationship between the GNSS transmitter, the reflecting surface and the receiver. Even small errors in the satellite’s propagated state shifts the estimated specular point by kilometers. Such errors can lead to misinterpretations of the delay-Doppler maps, for instance if the predicted specular point is over ocean when the true specular point occurs over land [gleason2020cygnss].

In real missions, the satellite orbit and relative geometry within satellite formations will naturally drift over time due to orbital perturbations. During mission design, simulation tools are used to predict the satellites motion. If the simulation does not accurately model the relevant perturbations, its predicted state evolution will diverge from physically realistic behavior, eventually yielding incorrect specular point locations and misleading conclusions about the performance of the GNSS-R system. Furthermore, inaccuracies in position and velocity directly translates to uncertainty in later analyses, like coverage analysis, or formation-keeping delta-V estimation. For these reasons, a high-fidelity propagator is necessary to ensure realistic and reliable evaluations of the GNSS-R system performance.

2.2 Problem Definition

A system simulator has already been developed to support the development of SmallSat Lab’s GNSS-R project (cite???). The tool currently allows area coverage estimation and a detection probability analysis of a stationary maritime target, and is intended to be extended for ocean surveillance and GNSS Radio Frequency interference (RFI) detection in the future. As explored in the previous section, these analyses rely on accurately propagating both GNSS and GNSS-R satellites to minimize their uncertainties. Currently,

the simulator uses the Simplified General Perturbations 4 (SGP4) propagator through the Skyfield Python library for this purpose.

The SGP4 propagator has long been the industry standard for orbit prediction, and has subsequently been the subject of thorough evaluation. Several studies document that the propagator can exhibit decreasing accuracy over longer time-horizons, particularly for formation-flight or precision geometry applications [aida2013sgp4_accuracy]. This raises concerns about whether the current SGP4-based implementation is sufficiently accurate for GNSS-R studies, which are highly sensitive to orbital state errors.

2.2.1 Research Objectives

This motivates a systematic verification of the Skyfield-SGP4 implementation by comparing its long-term predictions against a propagator with higher modeling fidelity. Therefore, this specialization project will support the SmallSat Lab GNSS-R mission by developing such a propagator within the Basilisk simulation framework, which natively supports detailed perturbation models and allows for controlled comparisons with Skyfield-SGP4. The study uses this capability to evaluate the differences between the propagators and evaluate their implications for GNSS-R mission analysis. As a result, the project is guided by three research questions:

- a

Answering these questions will support the project’s goal of guaranteeing the reliability and accuracy of orbital propagation tools used for GNSS-R small-satellite formation studies in future mission design.

2.3 Overall Method

2.4 Scope & Limitations

This specialization project focuses on evaluating the long-term orbital propagation accuracy of two distinct simulation frameworks, Skyfield-SGP4 and a high-fidelity Basilisk propagator, in the context of GNSS-R formation-flying missions. To ensure a controlled and way defined comparison, the project’s scope has been limited to a set of simulation configurations and perturbation models.

2.4.1 Simulation Configuration Scope

The analysis covers a relatively long time-horizon, ranging from several days up to one week, in order to adequately capture the frameworks’ divergence over a mission-relevant

time span. The simulated formation consists of two satellites flying in Low Earth Orbit (LEO) in a leader-follower configuration. Both are initialized on the same orbital plane with nominally constant along-track separation and with the same initial conditions to guarantee same-case comparisons.

2.4.2 Limitations

Several aspects of GNSS-R mission analysis lie outside the scope of this specialization project. First, no attitude dynamics are modeled. The satellites are assumed to maintain a fixed nadir-oriented attitude or otherwise idealized pointing, which removes attitude-orbit coupling and optical or RF pointing constraints from consideration.

Second, the project includes no GNSS-R signal modeling or bistatic radar simulation. The study evaluates orbital geometry only; signal processing elements such as delay-Doppler map generation, specular-point reflectivity, link budget modeling, or interference detection algorithms fall beyond the project's objectives.

Third, no active formation-keeping control is implemented. The satellites follow purely ballistic trajectories under the applied perturbations. As such, the delta-V analysis is limited to estimating how often and how strongly formation drift would need to be corrected, rather than simulating actual guidance, navigation, and control (GNC) actions.

Finally, conclusions drawn from this study are limited to the tested scenarios. The orbital regimes, perturbation models, and formation geometry considered here may not generalize to all LEO missions, nor to GNSS-R concepts employing different formation layouts, attitude profiles, or operational constraints.

These limitations naturally define the boundary between the present specialization project and a potential master's thesis. While this project focuses on verifying orbital propagation fidelity, a master's thesis could extend the analysis by incorporating full attitude modeling, GNSS-R signal simulation, operational formation-keeping strategies, and end-to-end detection performance. Such expansions would build upon the validated propagation models developed here, enabling a more comprehensive assessment of GNSS-R formation-flying missions.

2.5 Report Structure

Following is a brief overview of chapters in this report, and what each of them cover.

LITERATURE REVIEW

The literature review conducted for this paper can be divided into three themes:

- GNSS-R Small-Satellite Missions
- Formation Flying and Formation Maintenance for Small Satellites
- Integrated Satellite Mission Simulation and Lifetime Analysis

3.1 GNSS-R Small-Satellite Missions

- Scientific evolution:
 1. GNSS as a signal of opportunity
 2. GNSS-R feasibility for different remote sensing applications. Bistatic radar principle
 3. Early demonstrations of GNSS-R remote sensing on aircraft mission
 4. Spacecraft demonstrations
 5. Validation of technology for different applications
 6. Expansion of GNSS-R Mission applications (include jamming/spoofing detection)
- Mapping useful GNSS-R missions after validation of technology at spacecraft altitudes
- Feasibility of cooperative swarms and their respective formation types w. benefits

Reflected GNSS signals were first discussed as a

3.2 Formation Flying and Formation Maintenance for Small Satellites

3.3 Integrated Satellite Mission Simulation and Lifetime Analysis

3.4 Research gaps and thesis contributions

BACKGROUND THEORY

4.1 Reference Frames

A reference frame $\mathcal{F}^{rf} : \{\mathcal{O}_{rf}; \hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\}$ is uniquely defined by three dextral orthogonal unit vectors $\{\hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\}$ and the origin placement $\mathcal{O}_{rf}$ [1]. This section will present the definitions for the most important reference frames used in the thesis. The reader should note that the individual reference frame subscripts are assigned to be consistent with the conventional Basilisk notation.

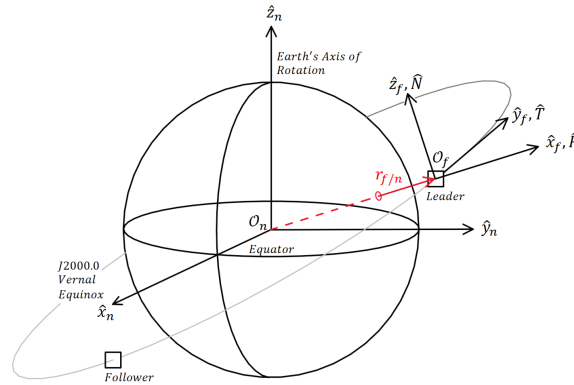


Figure 4.1.1: Illustration of the ECI ($\mathcal{F}^n$) and RTN ($\mathcal{F}^f$) reference frame definitions. The basis unit vectors are displayed with different lengths for illustrative purposes. Adopted from [2, 3] and modified from [4]

4.1.1 J2000 Earth-Centered Inertial (ECI) Frame

The ECI reference frame, $\mathcal{F}^n : \{\mathcal{O}_n; \hat{\mathbf{x}}_n, \hat{\mathbf{y}}_n, \hat{\mathbf{z}}_n\}$, is fixed relative to the distant stars, and is therefore commonly approximated as an inertial frame in which Newtonian mechanics are valid [5]. The frame is defined with its origin in the Earth's Center of Mass (CM), with $\hat{\mathbf{z}}_n$ pointing along its rotational axis towards the geographical North Pole, $\hat{\mathbf{x}}_n$ pointing towards the vernal equinox through the equator, and $\hat{\mathbf{y}}_n$ defined according to the right-hand rule. Multiple variants of the ECI definition exists, primarily varying in the epoch used to define the Earth's equatorial plane and equinox direction. In this thesis, the J2000 ECI frame is adopted. It is defined using the J2000.0 reference epoch, equivalent to 1st January 2000, at 12:00 Terrestrial Time (TT) [6]. The frame definition is illustrated in Figure 4.1.1.

4.1.2 Earth-Centered Earth-Fixed (ECEF) Frame

The Earth-Centered Earth-Fixed (ECEF) reference frame $\mathcal{F}^e : \{\mathcal{O}_e; \hat{\mathbf{x}}_e, \hat{\mathbf{y}}_e, \hat{\mathbf{z}}_e\}$ is fixed relative to the Earth's surface, with the frame origin positioned in its CM. This thesis adopts the WGS84 ECEF formulation as presented by Strykowski in [7], where the $\hat{\mathbf{x}}_e$ axis points through the Prime Meridian in the equatorial plane, $\hat{\mathbf{z}}_e = \hat{\mathbf{z}}_n$ points along the Earth's rotational axis, and $\hat{\mathbf{y}}_e$ is defined according to the right-hand rule. Because of the Earth's angular velocity, the transformation $\mathbf{R}_n^e$ between ECI to ECEF can be expressed using a time-dependent rotation angle $\theta(t)$ about the shared $\hat{\mathbf{z}}_e$ axis [4]

$$\mathbf{R}_n^e = \begin{bmatrix} \cos \theta(t) & -\sin \theta(t) & 0 \\ \sin \theta(t) & \cos \theta(t) & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{R}_e^n = (\mathbf{R}_n^e)^T \quad (4.1)$$

4.1.3 Radial-Transverse-Normal (RTN) Frame

The Radial-Transverse-Normal (RTN) reference frame, $\mathcal{F}^f : \{\mathcal{O}_f; \hat{\mathbf{x}}_f, \hat{\mathbf{y}}_f, \hat{\mathbf{z}}_f\}$, is a local orbital frame that is useful for expressing relative motion between spaceborne objects. Vallado presents the frame as the *Satellite coordinate system*, *RSW* in [8], but the *RTN* name is chosen for this thesis to better convey each component's physical significance. For use in formation flight, the coordinate frame origin is assigned to the leader satellite, l 's CM. The dextral orthogonal unit vectors are defined with the $\hat{\mathbf{x}}_f$ axis pointing in the radial direction along $\mathbf{r}_{l/n}^n$, the $\hat{\mathbf{z}}_f$ axis points normal to the orbital plane, and $\hat{\mathbf{y}}_f$ fulfills the right-hand rule. Note that the latter will always point in direction of positive velocity, but will only be perpendicular to the velocity for circular orbits [8, 9]. Using this definition, the frame can be expressed mathematically as:

$$\hat{\mathbf{R}}^n = \hat{\mathbf{x}}_f^n = \frac{\mathbf{r}_{l/n}^n}{\|\mathbf{r}_{l/n}^n\|}, \quad \hat{\mathbf{N}}^n = \hat{\mathbf{z}}_f^n = \frac{\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n}{\|\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n\|}, \quad \hat{\mathbf{T}}^n = \hat{\mathbf{y}}_f^n = \hat{\mathbf{z}}_f^n \times \hat{\mathbf{x}}_f^n \quad (4.2)$$

Here, $\mathbf{r}_{l/n}^n$ and $\mathbf{v}_{l/n}^n$ are used to denote the Earth-relative leader position and velocity expressed in ECI, while $\hat{\mathbf{y}}_f$ and $\hat{\mathbf{z}}_f$ correspond to the along- and cross-track directions, respectively. An illustration of the leader satellite's position relative to $\mathcal{O}_n$ and its affixed RTN frame is shown in Figure 4.1.1.

4.1.4 Body-Fixed Frame

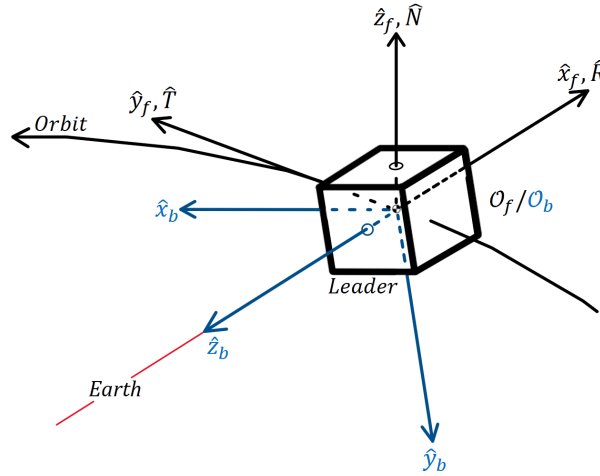


Figure 4.1.2: Illustration of the body-fixed ($\mathcal{F}^b$) and RTN ($\mathcal{F}^f$) reference frame definitions for a 1U CubeSat. Adopted from [10].

The body-fixed reference frame, $\mathcal{F}^b : \{\mathcal{O}_b; \hat{\mathbf{x}}_b, \hat{\mathbf{y}}_b, \hat{\mathbf{z}}_b\}$, is a moving reference frame that is rigidly attached to the spacecraft body. Because of this property, the body-fixed frame is used for expressing satellite attitude relative to an inertial frame, actuator torques and sensor measurements, since these quantities are naturally expressed relative to the body itself. The orientation of its axes are typically chosen to best represent the spacecraft geometry, and to coincide with the principal axes of inertia [11]. Consequentially, CubeSat body axes are commonly selected to be perpendicular to its side panels. The frame origin is commonly placed at the CM in order to simplify the rotational equations of motion, and to reduce translational-rotational coupling [12]. An illustration of the body frame definition for a 1U CubeSat is shown in Figure 4.1.2.

4.2 Modified Rodrigues parameters

Modified Rodrigues Parameters (MRP)s is a 3-element attitude parameterization commonly used for spacecraft attitude control. The representation can be derived from the Euler-Rodrigues symmetric parameters, better known as the unit quaternion:

$$\mathbf{q} = \begin{bmatrix} \epsilon \\ \eta \end{bmatrix} = [\epsilon_1 \quad \epsilon_2 \quad \epsilon_3 \quad \eta]^T, \quad \mathbf{q}^T \mathbf{q} = |\epsilon|^2 + \eta^2 = 1 \quad (4.3)$$

where η is the scalar component and ϵ is the imaginary vector. Using this definition, Shuster writes in [13] that the MRPs are obtained through a stereographic projection of the unit quaternion onto $\mathbb{R}^3$

$$\mathbf{p} = \frac{\epsilon}{1 + \eta}, \quad \mathbf{m} = \frac{\epsilon}{1 - \eta} \quad (4.4)$$

Here, $\mathbf{p}$ and $\mathbf{m}$ are the positive and negative forms, differing only in the singularity placement. This thesis utilizes the positive form, and will hereby be referred to as $\boldsymbol{\sigma}$. Using the axis-angle representation of a rotation, where $\hat{\mathbf{n}}$ represents the unit rotation axis and θ is the angle of rotation about the axis, the positive-form MRPs can be expressed as:

$$\mathbf{q} = \begin{bmatrix} \hat{\mathbf{n}} \sin(\theta/2) \\ \cos(\theta/2) \end{bmatrix} \Rightarrow \boldsymbol{\sigma}(\theta, \hat{\mathbf{n}}) = \hat{\mathbf{n}} \tan\left(\frac{\theta}{4}\right), \quad (4.5)$$

From this formulation, it follows that the MRPs become singular as $\theta \rightarrow 2\pi$, since

$$\tan\left(\frac{\theta}{4}\right) \rightarrow \infty \quad \text{for} \quad \theta \rightarrow 2\pi \quad (4.6)$$

To avoid numerical issues near the singularity, an equivalent MRP shadow set, $\boldsymbol{\sigma}^s$, may be used [14]. The shadow set transformation is defined as:

$$\boldsymbol{\sigma}^s = -\frac{\boldsymbol{\sigma}}{\boldsymbol{\sigma}^T \boldsymbol{\sigma}} \quad (4.7)$$

Switching to the shadow set when $\|\boldsymbol{\sigma}\| > 1$ effectively bounds the MRP set by the unit sphere, and prevents the parameterization from approaching the singularity numerically [15, 14, 13].

4.3 Classic Orbital elements

The classic orbital elements, also known as Keplerian elements, are a set of six parameters used to uniquely define the shape, size and orientation of an orbital plane, as well as the position of an orbiting body [16, 8]. In astrodynamics, these parameters provide a compact and intuitive representation of orbital motion, commonly used for orbital analyses and spacecraft guidance. The classic orbital elements used in this thesis are:

a : semimajor axis	}	orbital plane size and shape
e : eccentricity		
i : inclination	}	orbital plane orientation
Ω : right ascension of the ascending node (RAAN)		
ω : argument of perigee	}	orbital body position
M : mean anomaly		

It is important to note that the semimajor axis, and mean anomaly are in some orbital element configurations replaced with the angular momentum h , and the true anomaly θ , respectively. In any case, these six parameters define a Keplerian orbit, and they can be grouped according to the orbital property they describe: The semimajor axis and eccentricity define the shape and size of the orbital plane, while the inclination, Right Ascension of the Ascending Node (RAAN) and argument of perigee describe the plane's orientation. Finally, the mean anomaly is used to describe the spaceborne object's position along the orbital trajectory. An illustration of the classic orbital elements is shown in Figure 4.3.1.

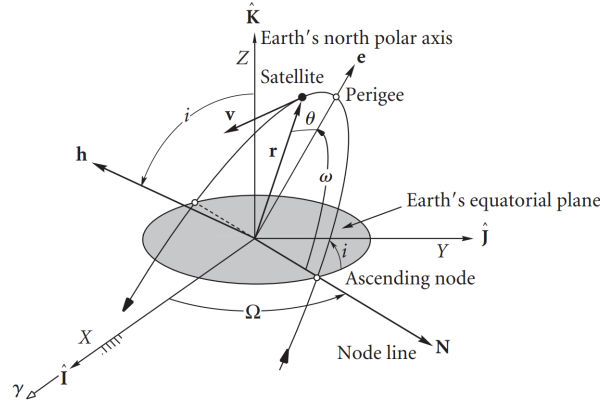


Figure 4.3.1: Orbital elements relative to the ECI frame. Courtesy of [16]

The semimajor axis is defined as half of the longest ellipse diameter, and determines the overall size of the orbit. For elliptical orbits, this parameter is directly coupled with the orbital period through Kepler's third law. The eccentricity is a measure of how much the orbital trajectory deviates from a circle. It takes values in the span between zero and one, where an eccentricity of zero corresponds to a perfectly circular orbit, and an increasing eccentricity correspond to an elliptical orbit of increasing elongation.

The inclination is the intersection angle between the orbital plane and the reference plane, commonly the equatorial plane spanned by the $\hat{x}_n$ and $\hat{y}_n$ axes in ECI. This angle determines the tilt of the orbital plane, while its orientation about the $\hat{z}_n$ axis is determined by the RAAN parameter. For nonzero inclinations, the orbital trajectory intersects the reference plane at two points, the ascending and descending nodes. The ascending node is where the object travels from the Southern to the Northern Hemisphere, and RAAN is defined as the angle between this node and the $\hat{x}_n$ axis in the equatorial plane.

The argument of perigee specifies the orientation of the ellipse within the orbital plane. It is defined as the angle from the ascending node to the perigee along the orbital trajectory, where perigee is a term specific to Earth satellites, describing the point along the trajectory closest to Earth. Finally, the mean anomaly defines the orbiting body's position along the orbit as a function of time. It is defined as the elapsed fraction of the orbital period since the orbiting body passed the perigee [16, 17].

4.4 Orbital Perturbations

D.A. Vallado defines orbital perturbations as deviations from the ideal Keplerian two-body motion caused by additional gravitational or non-gravitational accelerations acting on a spacecraft [8]. For an Earth-orbiting satellite, the Earth's oblateness, atmospheric drag, Solar Radiation Pressure (SRP) and third-body attraction from the Sun and Moon all cause its trajectory to deviate from the ideal point-mass solution. The following subsections will therefore address each major perturbation effect in approximate decreasing order of significance for LEO satellites in accordance with Montenbruck & Gill in [18].

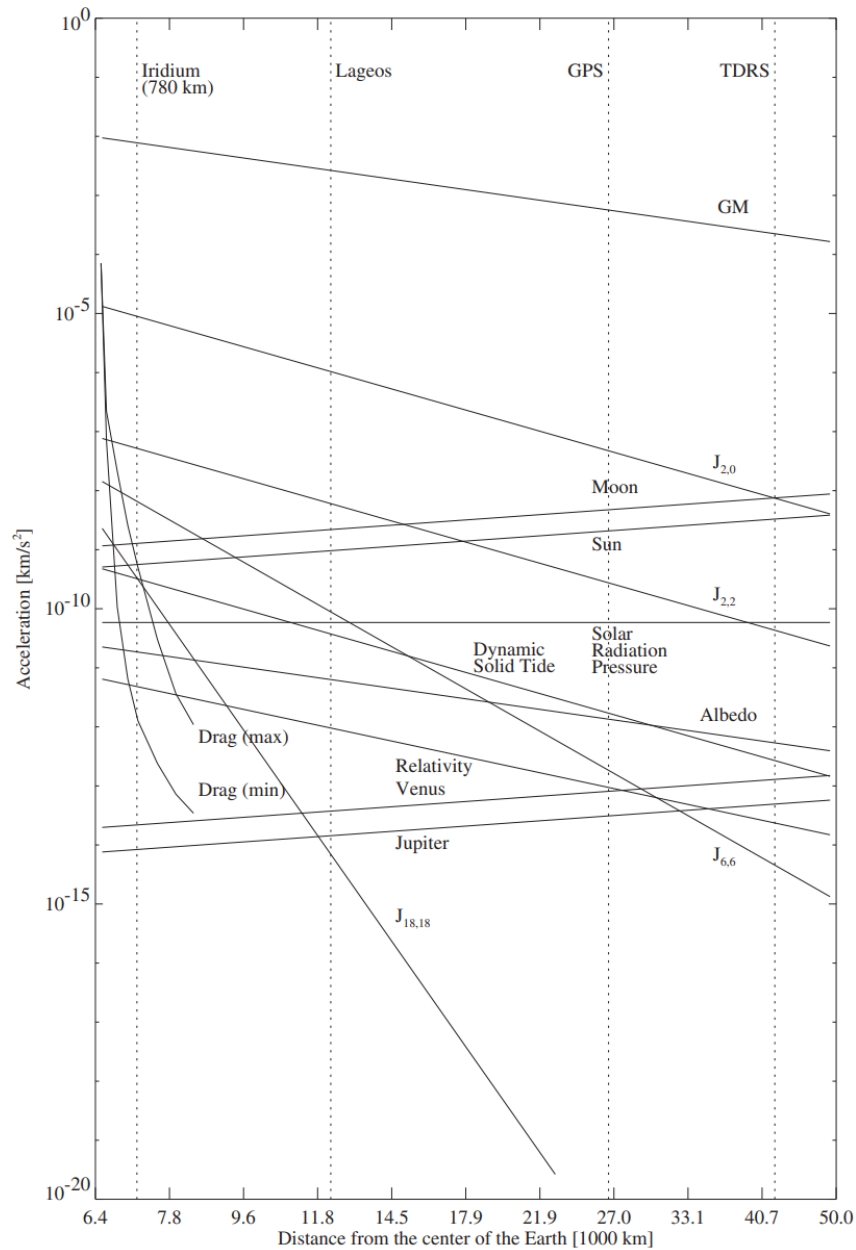


Figure 4.4.1: Acceleration magnitudes caused by various orbital perturbations. Courtesy of [18]

4.4.1 Earth's Oblateness

Keplerian motion assumes that the Earth can be represented as a perfectly spherical body with all its mass concentrated in the earth-centered inertial frame origin. Under this assumption, the gravitational force acting on a spacecraft, s , can be modeled using Newton's law of gravity

$$\mathbf{a}_{grav} = (\ddot{\mathbf{r}}_{s/n})_{grav} = -\frac{\mu}{\|\mathbf{r}_{s/n}\|_2^3} \mathbf{r}_{s/n} \iff (\ddot{\mathbf{r}}_{s/n})_{grav} = \nabla U, \quad \text{where } U = \mu \frac{1}{\|\mathbf{r}_{s/n}\|_2} \quad (4.8)$$

Using the notation from [4], $\mu = GM$ is the Earth's gravitational constant, and ∇U is the gradient of the gravity potential. In accordance with the spherical assumption, the gravity potential is only dependent on the satellite's geocentric distance, implying uniform density. In actuality, the Earth's mass is unhomogeneously distributed through its volume, and centrifugal forces causes the planet to bulge outwards around the equator due to its angular velocity. These irregularities produce perturbing accelerations that gradually alter the orbital trajectories over time.

Montenbruck & Gill explain that these deviations from spherical symmetry can be modeled by expanding the Earth's gravity potential into a spherical harmonic series representation [18]. The resulting series expresses the geopotential as a weighted sum of spherical harmonic terms, describing gravitational irregularities caused by the Earth's unhomogeneous mass distribution. Here, the spacecraft position is represented in ECEF spherical coordinates consisting of the geocentric distance $r = \|\mathbf{r}_{s/n}^e\|_2$, the geocentric latitude ϕ^e , and the longitude λ^e . Using these coordinates together with the associated Legendre polynomials, the geopotential may be written in the general spherical harmonic form as

$$U(r, \phi^e, \lambda^e) = \frac{\mu}{R} \sum_{n=0}^{\infty} \sum_{m=0}^n (C_{nm} V_{nm} + S_{nm} W_{nm}) \quad (4.9)$$

where,

$$V_{nm} = \left(\frac{R}{r}\right)^{n+1} P_{nm}(\sin \phi^e) \cos(m\lambda^e), \quad W_{nm} = \left(\frac{R}{r}\right)^{n+1} P_{nm}(\sin \phi^e) \sin(m\lambda^e). \quad (4.10)$$

Here, R denotes the Earth's radius, P_{nm} are the associated Legendre polynomials, while C_{nm} and S_{nm} are the normalized geopotential coefficients that encode the Earth's non-uniform mass distribution. The nm subscript denotes the degree n and order m , where increasing value correspond to decreasingly significant perturbation terms. The harmonic degree relates to longitudinal mass variation, while the order determines the longitudinal dependency on the harmonic term.

The reader should note that a common simplification is to assume order $m = 0$ due to comparatively small variations in the Earth's gravity field along the longitude. This yields the zonal harmonic expression, where the S_{n0} terms vanish, and the geopotential coefficients are commonly expressed as $J_n = -C_{n0}$. This gives the following expression derived from [18, 8]:

$$U(r, \phi^e) = \frac{\mu}{r} \left[1 - \sum_{n=2}^4 J_n \left(\frac{R}{r}\right)^n P_n(\sin \phi^e) \right] \quad (4.11)$$

The resulting zonal model captures the dominant long-term perturbation effects while significantly reducing the complexity of the gravity model. Amongst the zonal coefficients, the second degree term J_2 dominates the non-spherical effects, as is clearly evident by the substantially larger acceleration magnitude of $J_{2,0}$ in Figure 4.4.1 compared to the other terms. The Figure also illustrates that the higher degree and order terms rapidly decrease in magnitude. However, to include the dominant lateral and longitudinal non-spherical effect, this thesis employs the general spherical harmonic formulation in Equation 4.9, with truncated degree and order.

4.4.2 Atmospheric Drag

Orbital atmospheric drag is caused by a momentum exchange between the orbiting body's surface and colliding atmospheric molecules along its trajectory. According to Montenbruck & Gill, this effect is the dominant non-gravitational perturbation acting on a spacecraft in LEO, as is also illustrated in Figure 4.4.1. Since the drag force acts opposite to the spacecraft's velocity, it continuously removes orbital energy, and causes a gradual reduction in orbital altitude and velocity over time [8, 18, 4].

For spacecraft operating in LEO, the interaction between the spacecraft surface and surrounding atmosphere is commonly modeled using the free-molecular flow assumption. In this regime, the atmospheric density is sufficiently low such that the inter-molecular collisions are negligible compared to the molecule-surface interaction. The drag acceleration experienced by a spacecraft s can then be expressed using the classical aerodynamic formulation, as presented in [18]

$$\mathbf{a}_D^n = (\ddot{\mathbf{r}}_{s/n}^n)_D = -\frac{1}{2}C_D \frac{A_D}{m_s} \rho v_{rel}^2 \hat{\mathbf{v}}_{rel}^n \quad (4.12)$$

where C_D is the dimensionless drag coefficient, A_D is the effective cross-sectional area normal to the incoming flow, m_s is the spacecraft mass and ρ is the local atmospheric density. v_{rel} denotes the relative speed, while $\hat{\mathbf{v}}_{rel}^n$ is the unit vector in the direction of relative velocity. It is important to note that the relative velocity differs from the inertial spacecraft velocity because it represents its velocity relative to the local atmosphere [4]. Following Montenbruck & Gill in [18], the relative velocity can therefore be approximated by assuming the atmosphere co-rotates with the Earth

$$\mathbf{v}_{rel}^n = \mathbf{v}_{s/n}^n - \boldsymbol{\omega}_{e/n}^n \times \mathbf{r}_{s/n}^n \quad (4.13)$$

where $\{\mathbf{r}_{s/n}^n, \mathbf{v}_{s/n}^n\}$ are the spacecraft position and velocity expressed in the ECI frame, and $\boldsymbol{\omega}_{e/n}^n$ is the Earth's angular velocity vector. Amongst the parameters in the classic drag formulation, the atmospheric density ρ is generally the largest source of modeling uncertainty. In this thesis, two density models will be evaluated: the NRLMSISE-00 model and the exponential density model. The exponential density model is a heavily simplified alternative, and assumes exponentially decreasing density with increasing altitude.

$$\rho(h) = \rho_0 \exp\left(-\frac{h - h_0}{H_0}\right), \quad H_0 = \frac{\mathcal{R}T}{\mu_m g} \quad (4.14)$$

Here, ρ_0 is the reference density at some geocentric reference height h_0 , and h is the spacecraft altitude above the reference height. Finally, H_0 is the density scale height, whose

equation in 4.14 can be derived from the hydrostatic equation and gas law from thermodynamics. This model captures the approximate atmospheric decay with altitude, but fails to account for major spatial and temporal effects, such as solar activity, geomagnetic disturbances, longitudinal/lateral variations and long-term solar cycles [18].

The NRLMSISE-00 empirical model accounts for these spatial and temporal atmospheric variations, and has been developed using several decades of atmospheric observations and satellite measurements. In addition to the density, the model also estimates other atmospheric parameters, such as temperature and individual molecule densities. Because of the spatial and temporal dependency of these parameters, the empirical model requires additional inputs describing both the spacecraft environment and current atmosphere conditions. These include the spacecraft altitude, like the exponential density model, but also the geographic location, current time and different time-averaged measurements of solar activity represented through the $F_{10.7}$ solar radio flux, and the geomagnetic index A_p [19]. Together, these parameters allow the model to account for short- and long-term variations in the atmospheric density.

4.4.3 Solar radiation pressure

Like atmospheric drag, SRP is a non-gravitational perturbation caused by momentum transfer between the spacecraft and incident particles. However, in the case of SRP, this transfer originates from photons rather than residual atmospheric molecules. When photons are absorbed or reflected by the spacecraft surface, they transfer their momentum to the spacecraft, generating a small force in the approximate direction away from the Sun. Because the force acts continuously over the illuminated surface area, it is referred to as a radiation pressure [18, 4].

As discussed by Montenbruck & Gill in [18], the radiation pressure does not vary with the geocentric altitude directly, but rather the distance from the Sun according to the inverse-square law. Because the variation in Sun-spacecraft distance is negligible compared to the Earth orbital altitude, the resulting SRP acceleration experienced by an Earth-orbiting spacecraft is approximately constant, as illustrated in Figure 4.4.1. Although the SRP acceleration remains approximately constant with altitude, its relative significance increases at higher altitudes where atmospheric drag becomes progressively weaker, especially for spacecraft with large area-to-mass ratios.

For a flat spacecraft surface, the resulting SRP acceleration depends on the relative geometry between the solar radiation direction $\mathbf{r}_{Sun/s}^n$, and the spacecraft surface normal $\hat{\mathbf{n}}$. Using the formulation presented by Montenbruck & Gill, the acceleration in the ECI frame can be written as

$$\mathbf{a}_R^n = (\ddot{\mathbf{r}}_{s/n}^n)_R = -P_{Sun} \frac{AU^2}{\|\mathbf{r}_{Sun/s}^n\|_2^2} \frac{A_R}{m_s} \cos(\theta) [(1 - \varepsilon)\mathbf{r}_{Sun/s}^n + 2\varepsilon \cos(\theta)\hat{\mathbf{n}}_s] \quad (4.15)$$

where P_{Sun} denotes the solar radiation pressure at one astronomical unit (AU), the parameter A_R/m_s represents the spacecraft sun-normal cross-sectional area-to-mass ratio, and θ is the angle between the incident radiation direction and $\hat{\mathbf{n}}$. Finally, ε denotes to the fraction of reflected incoming radiation, where $\varepsilon = 0$ and $\varepsilon = 1$ correspond to perfect absorption and reflection, respectively.

For many orbit propagation applications, Montenbruck & Gill further argue that the simplified assumption $\theta = 0$ provides sufficient accuracy [18]. Under this assumption, the SRP model reduces to the commonly used cannonball model

$$\mathbf{a}_R^n = (\ddot{\mathbf{r}}_{s/n}^n)_R = -P_{Sun} C_R \frac{A_R}{m_s} \frac{\mathbf{r}_{Sun/s}^n}{\|\mathbf{r}_{Sun/s}^n\|_2^3} \text{AU}^2 \quad (4.16)$$

where C_R is the radiation pressure coefficient. This approximation models the spacecraft as an idealized spherical body with uniform reflective properties, thus simplifying the SRP model while keeping the dominant perturbation behaviour [8, 4].

4.4.4 Third-Body Perturbations

In addition to the Earth's gravitational field, Earth-orbiting spacecraft are subjected to gravitational attraction from other massive bodies within the solar system. For satellites orbiting around the Earth, the Sun and Moon are the dominant sources of third-body perturbation. Although these bodies are located far from the spacecraft compared to the Earth, their gravitational influence still produce measurable orbital perturbations, particularly for satellites operating in Medium Earth Orbit (MEO), Geostationary Earth Orbit (GEO), or highly elliptical trajectories, as presented by Figure 4.4.1 [18, 4].

Third-body perturbations introduce additional gravitational accelerations that continuously vary with the relative geometry between the spacecraft, Earth and the perturbing body. Although the perturbing accelerations from the Sun and Moon are much smaller than the Earth's central gravitational attraction, Montenbruck & Gill note that the influence become increasingly significant with orbital altitude, becoming comparable to geopotential perturbations of low degree [18]. Because the perturbing acceleration changes direction throughout the orbit, the resulting effect manifests as gradual deviations from ideal Keplerian motion, and contributes both periodic and long-term variations in the spacecraft trajectory, which is consistent with the general three-body dynamics described by Curtis in [16].

Following the formulation presented by Montenbruck & Gill, the perturbing acceleration from a third massive body B is obtained by subtracting the gravitational acceleration exerted by the perturbing body on the Earth from the gravitational acceleration acting on the spacecraft. Using the same notation as in [4], the perturbing acceleration expressed in the ECI frame becomes

$$\mathbf{a}_B^n = (\ddot{\mathbf{r}}_{s/n}^n)_B = \mu_B \left(\frac{\mathbf{r}_{B/n}^n - \mathbf{r}_{s/n}^n}{\|\mathbf{r}_{B/n}^n - \mathbf{r}_{s/n}^n\|_2^3} - \frac{\mathbf{r}_{B/n}^n}{\|\mathbf{r}_{B/n}^n\|_2^3} \right) = \mu_B \left(\frac{\mathbf{r}_{B/s}^n}{\|\mathbf{r}_{B/s}^n\|_2^3} - \frac{\mathbf{r}_{B/n}^n}{\|\mathbf{r}_{B/n}^n\|_2^3} \right) \quad (4.17)$$

When multiple perturbing bodies are considered simultaneously, the total third-body acceleration is obtained by summing the individual contributions from each body. Using the major perturbing bodies for a LEO satellite, this becomes

$$\mathbf{a}_{B,\text{tot}}^n = \mathbf{a}_{Sun}^n + \mathbf{a}_{Moon}^n \quad (4.18)$$

4.5 Differential Perturbations in Formation-Flying

Alfriend et al. refers to the generally agreed upon definition of spacecraft formation flying in [20] as: "The tracking or maintenance of a desired relative separation, orientation or position between or among spacecraft". It then follows that differential perturbations describe the relative effect of environmental accelerations acting on two satellites in formation, that cause the formation to naturally drift away from its desired geometry over time. The difference in experienced accelerations by the two spacecraft in formation can arise because the spacecraft occupy slightly different orbits, positions, velocities and attitudes, or because they have different physical attributes. In formation-flying, even small prolonged differences in experienced acceleration can accumulate over many orbital periods to gradually change the radial, along- or cross-track separation. Differential perturbations therefore motivate the need for formation-control strategies to preserve the desired formation geometry within mission constraints. Since stronger natural drift generally requires more frequent or larger corrective measures, differential perturbations also become direct drivers for formation maintenance fuel consumption, and thus, also mission lifetime [18, 20]. The following subsections will therefore summarize the most important differential perturbation effects for LEO orbiting formations.

4.5.1 Natural Relative Drift Under Ideal Two-Body Motion

For a leader-follower spacecraft pair, the relative orbit can be described by the difference between their respective classic orbital elements. If the leader has orbital elements α_L , and the follower has α_F , the differential orbital elements can be expressed as

$$\Delta\alpha = \alpha_F - \alpha_L = [\delta a \quad \delta e \quad \delta i \quad \delta \Omega \quad \delta \omega \quad \delta M_0]^T \quad (4.19)$$

where the components represent the differences in semimajor axis, eccentricity, inclination, RAAN, argument of perigee and mean anomaly at a chosen epoch, respectively. Under the ideal two-body assumption, where the Earth is modeled as a spherical body and all perturbing forces from Section 4.4 are neglected, the only form of secular formation drift is caused by a difference in the semimajor axis. This is stated by Alfriend et al. in [20], who explain that the drift only manifest in the along-track direction, resulting in continuous differential mean anomaly growth. This interpretation is also consistent with Hammerl & Schaub, who show that the relative radial offset must be zero for bounded relative motion in [21].

Other differential orbital elements still contribute to natural relative motion, but they do not cause secular drift in the ideal two-body problem. Instead, they induce bounded periodic motion. For example, a differential inclination produces periodic cross-track motion as the two orbital planes are tilted with respect to each other. Similarly, differential eccentricity contributes to in-plane radial and along-track oscillations. These periodic relative motions may be important for the formation geometry, but they do not themselves cause the formation to drift apart. However, additional secular drift mechanisms arise from differential perturbations, especially differential J_2 effects and atmospheric drag [22].

4.5.2 Differential J_2 Perturbations

As presented in Section 4.4.1, J_2 is the second degree, zeroth order zonal coefficient in the spherical harmonic representation of the Earth's gravity field. The coefficient represents the dominant term associated with the Earth's oblateness, and the zonal model of the same degree and order is therefore commonly known as the J_2 perturbation model. In a formation-flying context, the differential J_2 perturbation refers to the difference between the J_2 -induced accelerations acting on the two spacecraft, and is one of the most important causes of secular formation drift in LEO.

The J_2 perturbing acceleration causes a slow precession of the orbit, seen primarily as precession of the orbital plane and a rotation in the argument of perigee. Alfried et al. gives the corresponding per-satellite secular rates for RAAN and the argument of perigee in [20], which can be expanded to the following expression [23]

$$\dot{\Omega} = -\frac{3\bar{n}J_2R^2}{2a^2(1-e^2)^{1/8}} \cos i \quad (4.20)$$

$$\dot{\omega} = -\frac{3\bar{n}J_2R^2}{2a^2(1-e^2)^{1/8}} \left(\frac{5}{2} \sin^2 i - 2 \right) \quad (4.21)$$

where $\bar{n}$ is the mean motion, R is the Earth radius, and a , e and i are the satellite semimajor axis, eccentricity and inclination, respectively. Note that these per-satellite equations can be expressed as a function of each differential orbital elements by Taylor series expansion about the leader orbit, which has been derived for differential inclination in [20]. Examining Equations 4.20 and 4.21 shows that a difference in either semimajor axis, eccentricity or inclination between the satellites in a formation can cause J_2 -induced secular rates, resulting in secular drift. This effect will manifest in the RTN frame as radial, cross- and along-track drift, depending on which secular rates differ. Because both secular rates are inversely proportional to $(1-e^2)$ a difference in eccentricity will cause much smaller secular rate than a corresponding difference in inclination for near-circular orbits where $e \approx 0$ [20].

4.5.3 Differential Atmospheric Drag

The atmospheric drag acceleration acting on a single spacecraft was introduced in Section 4.4.2. For formation-flying, however, the relative effect of drag is determined by the difference between the drag accelerations acting on each spacecraft. D'Amico & Montenbruck explain in [22], that atmospheric density variations can be neglected for close formations with sub-kilometer separation. Under this approximation, the main source of differential atmospheric drag is not a difference in local atmosphere, but a difference in the spacecraft ballistic coefficients. The ballistic coefficient is defined as

$$b = C_D \frac{A_D}{m_s} \quad (4.22)$$

where C_D , A_D and m_s are the drag coefficient, flow-normal cross-sectional area and spacecraft mass from equation 4.12. If the leader and follower spacecraft have ballistic coefficients b_L and b_F , the relative ballistic coefficient difference can be written as $\varepsilon = (b_F - b_L)/b_L$. The resulting relative along-track drift over a time duration Δt with a constant relative ballistic coefficient difference is given in [22]

$$\Delta r_{\hat{T}} = \frac{1}{2} \varepsilon \|\mathbf{a}_D\|_2 \cdot \Delta t^2 \quad (4.23)$$

where $\|\mathbf{a}_D\|_2$ denotes the drag acceleration magnitude of the leader spacecraft. This expression shows that small ballistic coefficient differences can accumulate into substantial along-track drift if the difference is persistent throughout significant time. Finally, it is important to note that the ballistic coefficient is attitude dependent though the flow-normal cross-sectional area, which makes relative spacecraft attitude an important driver for relative along-track drift and subsequent fuel consumption [20, 22].

4.6 GNSS-R Mission Concepts

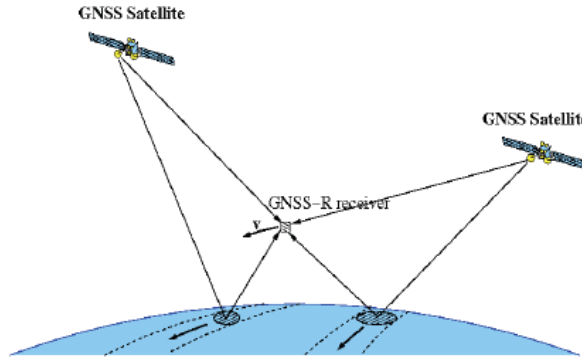


Figure 4.6.1: Illustration of the GNSS-R measurement principle. Courtesy of [24]

4.6.1 Remote Sensing Principles

GNSS-R is a passive remote sensing technique that uses reflected GNSS signals as signals of opportunity. As shown in Figure 4.6.1, GNSS-R forms a bistatic radar configuration where the GNSS satellite acts as a non-cooperative transmitter, while the observing spacecraft carries the receiver. Zavorotny et al. describes this concept in [25] as GNSS bistatic radar of opportunity, where the broadcasted GNSS signals illuminate the Earth's surface, and the receiver processes the scattered signal to infer surface properties such as ocean wave height, wind speeds, soil moisture, etc.

4.6.2 Cooperative GNSS-R Measurements

4.6.3 Operational Constraints

ALREADY REFERENCED in "Pointing Guidance System": Requires GNSS receiver at the zenith satellite, and GNSS-R receiver at the opposite, nadir, side

4.7 Basilisk Process-Task-Model Hierarchy

Basilisk simulations are constructed from a hierarchy of processes, tasks and models. A model is the lowest-level simulation object, and represents an individual computational component, such as a spacecraft dynamics model, environmental effector, guidance module, controller, actuator model or data recorder. In this thesis, models are grouped according to their functional role. Environmental models describe the spatial conditions, such as gravity bodies, atmospheric density, eclipse conditions and ground-station access. Dynamics models describe the physical spacecraft and its interacting forces and subsystems, including translational and attitude dynamics, perturbation effectors and actuators. Flight software models describe onboard Guidance Navigation and Control (GNC) algorithms.

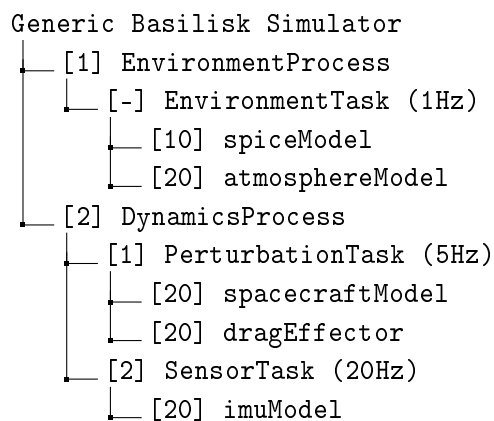


Figure 4.7.1: Hierarchical process-task-model structure of a generic Basilisk simulator. The square brackets denote the assigned object priority, and respective task update rates are displayed in the parentheses

Models are scheduled for execution by assigning them to tasks, where each task has a specified update rate. Tasks are in turn assigned to processes, which provide a higher-level grouping of related tasks. An illustration of a process-task-model hierarchical structure for a generic Basilisk simulator is shown in Figure 4.7.1. The execution order is determined by the assigned priority, where higher-order objects are executed first. Same-priority objects or object without a specified priority are updated according to their initialization order. Using these rules, the execution order of the generic Basilisk simulator would thus be:

`imuModel` $\rightarrow$ `spacecraftModel` $\rightarrow$ `dragEffector` $\rightarrow$ `atmosphereModel` $\rightarrow$ `spiceModel`

COMPARATIVE PROPAGATION SIMULATOR

The Comparative Propagation Simulator developed in this thesis is a continuation of the comparative propagation framework from the specialization project [4]. In that work, a Basilisk numerical propagator was compared against a Skyfield SGP4 propagation baseline to assess the impact of individual perturbation effects and numerical integrator configuration to evaluate Basilisk long-horizon prediction accuracy. However, two important limitations motivated further development of the simulator. First, the Skyfield SGP4 baseline was itself subject to prediction inaccuracies, which made it a weak reference for assessing long-horizon prediction accuracy. Two-Line Element (TLE)-based propagation, such as SGP4, is primarily intended for efficient near-epoch orbit prediction, and is known to diverge over longer propagation intervals. Second, the Basilisk exponential atmospheric model used in the specialization project was too simple and wrongly tuned, which resulted in significantly smaller drag accelerations than expected.

The Comparative Propagation Simulator developed for this thesis aims to address these limitations by extending both the Skyfield and Basilisk parts of the comparison. The following sections present the implementation of the simulator at a high level of abstraction to provide an understanding of how the separate propagators are initialized, and how the output data is generated and made comparable. First, the overall simulator architecture is introduced, including the configuration system, data flow, execution sequence, and output files. Then the simulator input parameters will be presented in further detail. Specific parameters or simulation configuration is outside the scope of this chapter, but are presented in 7. For future reference, the term *simulator* will be used to address the Comparative Propagation Simulator in its entirety, while *sub-simulation* will be used to denote the individual Skyfield or Basilisk propagators.

5.1 Simulation Architecture

The overall structure and execution order of the Comparative Propagation Simulator can be represented as a sequence of three main program blocks: initialization of the simulation environment, Skyfield SGP4 propagation, and Basilisk numerical propagation. This execution order is necessary because the Basilisk sub-simulation is initialized using the Skyfield-propagated spacecraft states at the simulation start time. A more detailed overview of the simulator architecture is shown in Figure 5.1.1. The figure illustrates the execution sequence from left to right, as well as the abstracted program flow within each program block. The vertically stacked layers separate file inputs, internal object instances, program blocks and output files. The top layer contains the user-defined input files, including configuration files, TLE data, and external model data. The middle layer shows the main program blocks, while the adjacent layers show how object instances and data are created and passed between blocks. Finally, the bottom layer shows the output files generated by the simulator.

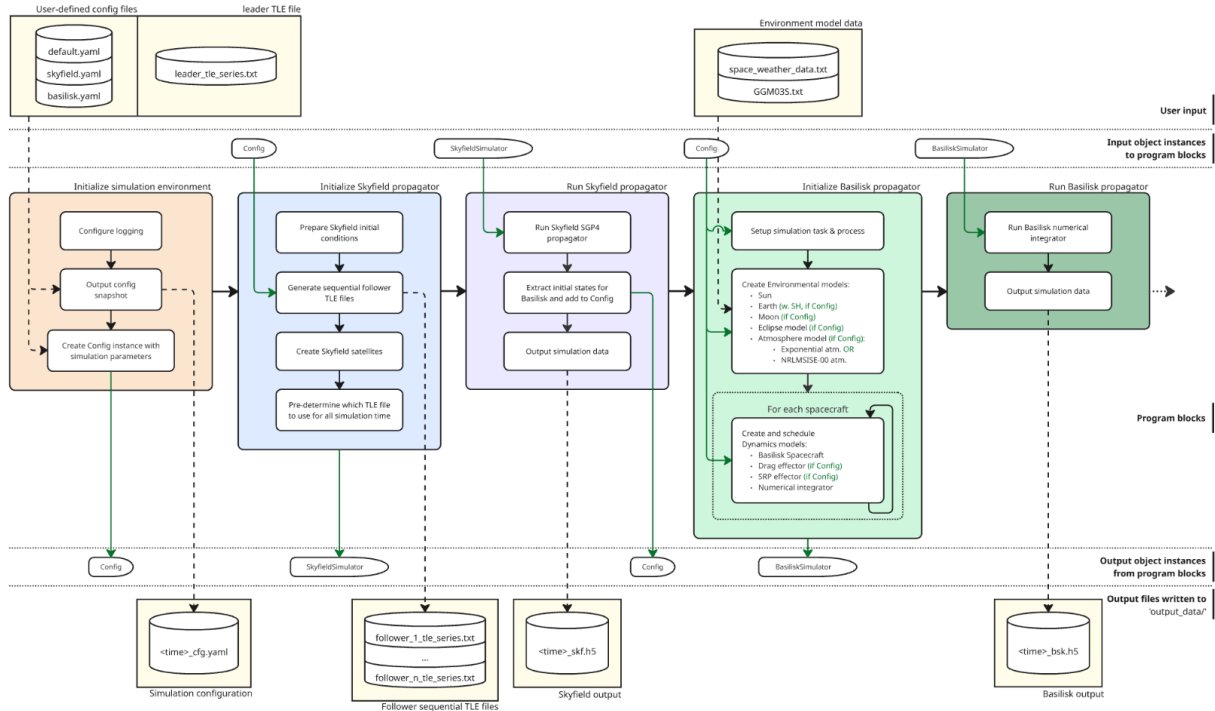


Figure 5.1.1: Detailed Comparative Propagation Simulator flow diagram. Adopted from [4]. A full-size version of this diagram is shown in the Appendix Figure .0.2

5.1.1 Initialization of Propagator Simulation Environment

When the simulator is executed, the simulation environment is initialized by loading the configuration files and the leader TLE time series. The file `default.yaml` defines the parameters shared by both propagation methods, while `skyfield.yaml` and `basilisk.yaml` define settings specific to the Skyfield and Basilisk sub-simulations, respectively. The file `leader_tle_series.txt` contains a sequence of TLEs for the leader spacecraft, ordered by epoch, spanning the full simulation duration with a maximum epoch separation of 12

hours. The use of this TLE time series is described in more detail in Section ???. For traceability, a timestamped snapshot of the combined simulation configuration is written to file, allowing the exact configuration used to generate a given set of outputs to be inspected later. The validated configuration parameters are then collected into a shared configuration instance that is used by the subsequent Skyfield and Basilisk program blocks.

5.1.2 Skyfield SGP4 Propagator Flow

The Skyfield SGP4 sub-simulation is initialized after the shared configuration instance has been created. During this step, the leader TLE time series is loaded and used to construct the corresponding follower TLE time series with constant shifted mean anomaly relative to the leader. These are then used to create Skyfield satellite objects, which define the SGP4 propagation model used for each spacecraft. Finally, the simulator pre-determines which specific TLE from the time series should be used at any given simulation time. This produces a time-indexed mapping between simulation time and TLE set, allowing the subsequent Skyfield propagation block to evaluate each spacecraft state using the TLE closest to the requested propagation time.

After the Skyfield propagator has been initialized, the Skyfield SGP4 sub-simulation is executed for the full simulation duration. At each simulation time step, the simulator selects the pre-determined TLE for each spacecraft, and evaluates the corresponding SGP4 model to obtain the spacecraft position and velocity in the ECI frame. This produces a time series of translational states for both the leader and follower spacecraft, where the propagation remains tied to the continuously updated TLE series rather than a single fixed TLE epoch. When the Skyfield propagation is complete, the states at the initial simulation time are extracted, and the resulting state time series are written to a timestamped output file, `<time>_skf.h5`. This output represents the Skyfield comparison baseline used in the later comparison against the Basilisk numerical propagation.

5.1.3 Basilisk Propagator Flow

Once the Skyfield SGP4 sub-simulation has completed, the Basilisk propagator is initialized. The initialization first creates the Basilisk simulation process and task, which are required by the Basilisk framework to schedule and integrate all attached simulation models. The environmental and spacecraft dynamics models are then created according to the selected simulation configuration. In contrast to the Skyfield sub-simulation, where the propagation behavior is determined by the selected TLEs and the SGP4 model, the Basilisk sub-simulation allows different combinations of explicitly modeled perturbation effects to be enabled or disabled. This configurability is indicated by the (*if config*) annotations in Figure 5.1.1, and makes it possible to run otherwise identical propagation cases with different environmental model combinations. Finally, the leader and follower spacecraft are initialized and the numerical integration method is configured.

After initialization, the Basilisk numerical propagation is run for the entire simulation duration. During this process, Basilisk integrates the spacecraft equations of motion using the configured numerical integration method and the environmental and dynamics

models attached during initialization. When the propagation is complete, the translational states of each spacecraft is written to a timestamped output file, `<time>_bsk.h5`. This completes the Comparative Propagation Simulator execution and provides the Basilisk state time series used for comparison against the Skyfield SGP4 baseline.

5.2 Simulator Inputs

As mentioned in Section 5.1.1, each simulation case is defined through three user-defined configuration files, which have been summarized in Table 5.2.1. This structure makes it easy to keep the simulated scenario fixed while independently changing the configuration of each propagation method. The file `default.yaml` contain all parameters required to define the simulation scenario. This includes the simulation start time and duration, the leader TLE time series, the number of satellites, the initial in-plane separation between the leader and follower, spacecraft physical parameters, and plotting options.

Config file	Input parameters	Unit
default.yaml	Simulation start time	UTC
	Simulation duration	h
	Leader TLE	-
	In-plane separation angle	deg
	Number of satellites	-
	For each satellite:	-
	Mass	kg
	Drag coefficient	-
	Cross-section area perpendicular to velocity	m ²
	Radiation pressure coefficient	-
	Cross-section area perpendicular to the Sun-vector	m ²
skyfield.yaml	Data processing and plotting flags	-
	SGP4 step size	sec
basilisk.yaml	Basilisk step size	sec
	Numeric integrator	-
	Spherical harmonics degree	-
	Enable spherical harmonics flag	-
	Enable exponential density drag flag	-
	Enable MSIS drag flag	-
	Enable solar radiation pressure flag	-
	Enable Sun third-body pull flag	-
	Enable Moon third-body pull flag	-

Table 5.2.1: A complete list of all Comparative Propagation Simulator configuration input parameters. Adopted from [4]

The file `skyfield.yaml` contains the Skyfield-specific propagation setting, which is only the time step between each SGP4 evaluation. Finally, the `basilisk.yaml` file defines the Basilisk propagation setup, including the numerical integrator, integration step size,

spherical harmonics degree, and boolean flags for enabling or disabling individual perturbation models. In addition to the configuration files, the Basilisk sub-simulation uses external model data for selected environmental models. This includes the gravity coefficient file used by the spherical harmonics model and the space-weather data used by the NRLMSISE-00 atmosphere model.

GNSS-R FORMATION FLIGHT SIMULATOR

The GNSS-R Formation Flight Simulator is developed to estimate the operational lifetime of a realistic dual-satellite GNSS-R formation-flying mission. While the Comparative Propagation Simulator establishes confidence in the Basilisk translational propagation environment, the formation flight simulator extends this environment into a closed-loop mission simulation where orbital motion, attitude dynamics, power availability, actuator usage, and autonomous operational decisions are evaluated together. The following sections describe the simulator architecture and the implemented modules that enable mission lifetime analysis. These include an operational mode switching logic that selects spacecraft activities based on environmental, temporal, and battery constraints, an EPS model that tracks power generation, storage, and consumption, and the GNC functionality required for pointing and formation maintenance. The latter includes a pointing guidance system for generating attitude references from the active operational mode, an MRP feedback pointing controller for attitude tracking, and a formation controller for obtaining and maintaining the desired leader-follower geometry.

The implemented mission logic, EPS and GNC depend on the spacecraft components available to each satellite. This chapter therefore assumes that each spacecraft is equipped with a propulsion system consisting of a propellant tank and thruster, a three-axis controllable Reaction Wheel (RW) configuration, an On-Board Computer (OBC), solar panels, a battery pack with a built-in heating element, a power distribution module, a radio communication antenna, and a payload consisting of a high-gain GNSS-R receiver and a lower-gain GNSS receiver. These components are introduced here only to clarify the subsystem models referenced throughout the chapter, while their exact hardware values and mounting configuration are described later in Section 7.1. Finally, throughout this chapter, the term simulator refers to one complete Monte Carlo (MC) campaign, consisting of multiple GNSS-R formation-flight simulation runs. Each simulation run corresponds to one complete Basilisk execution with a resolved set of configuration parameters.

6.1 Simulation Architecture

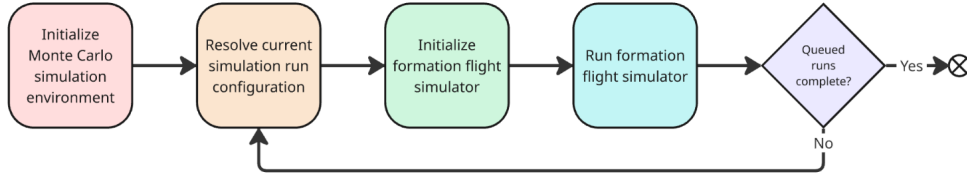


Figure 6.1.1: Simplified GNSS-R formation flight simulator flow diagram

The high-level execution flow of the GNSS-R Formation Flight Simulator is shown in Figure 6.1.1. The simulator is organized into five main program blocks: initialization of the MC simulation environment, resolution of the current simulation run configuration, initialization of the formation flight simulator, execution of the formation flight simulator, and evaluation of whether additional MC runs remain. These blocks can be divided into two main parts, separated by the vertical dashed line in the detailed flowchart in Figure 6.1.2. The left part acts as a simulation run orchestrator, where the MC environment is configured, and all run-specific configuration files are prepared. The right part represents the iterative execution of individual simulation runs, where each run is initialized, executed, and written to file before the next queued run is started. As in the Comparative Propagation Simulator architecture presented in Section 5.1, the detailed flowchart is divided into vertically stacked layers representing input files, internal object instances, program blocks, transferred output objects, and generated output files. However, unlike the Comparative Propagation Simulator, the GNSS-R Formation Flight Simulator does not include a separate Skyfield propagation module, but is built entirely upon the Basilisk framework.

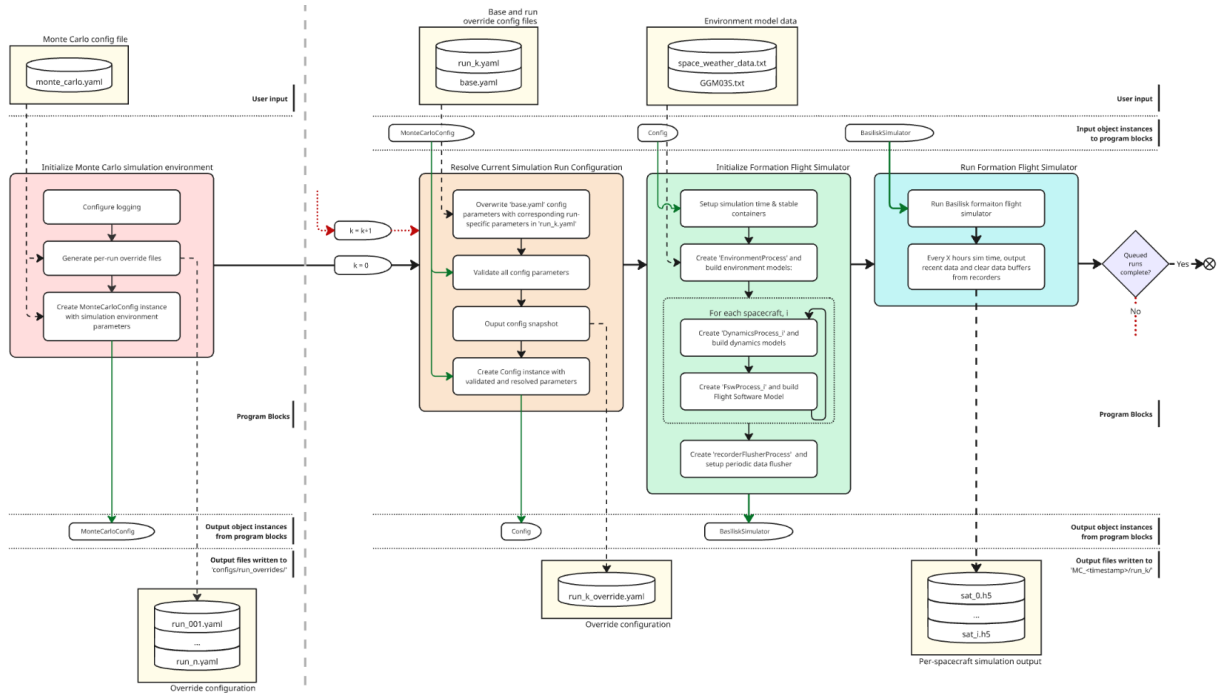


Figure 6.1.2: Detailed GNSS-R formation flight simulator flow diagram. A full-size version of this diagram is shown in the Appendix Figure .0.2

6.1.1 Monte Carlo Environment Initialization Flow

The first program block initializes the MC simulation environment from the user-defined configuration file `monte_carlo.yaml`, and is considered the simulation run orchestrator. In this thesis, MC execution is assumed to be enabled for all simulation cases, such that the simulator is structured around a queue of formation-flight simulation runs rather than a single isolated run. The MC configuration defines the number of queued runs and the run-specific parameter variations used to generate override configuration files. If n simulation runs are queued, $n - 1$ override files are generated. Finally, the validated MC settings are collected into a global MC configuration instance, which is passed to the iterative part of the simulator and determines how the queued formation-flight simulations are executed.

6.1.2 Current Run Configuration Resolution Flow

The next program block resolves the configuration for the current simulation run. This step is similar to the configuration initialization used in the Comparative Propagation Simulator, described in Section 5.1.1, but is extended to support iterative MC execution. For the first simulation run, the configuration is defined directly by the default scenario in `base.yaml`. For all subsequent runs, the corresponding override file `run_k.yaml` is loaded and used to replace selected parameters in `base.yaml`, where k denotes the current simulation run index. This allows each queued run to modify a limited set of scenario parameters while preserving the remaining default configuration. Once the current run configuration has been resolved, all parameters are validated and written to a timestamped configuration snapshot. As in the Comparative Propagation Simulator, this snapshot provides traceability by allowing each generated output data set to be linked to the exact parameter values used to produce it. Finally, the validated parameters are collected into a run-specific `Config` instance, which is passed to the formation flight simulator initialization block.

6.1.3 Single Formation Flight Initialization & Execution Flow

The resolved run configuration is then used to initialize and execute one complete Basilisk GNSS-R formation flight simulation. At a high level, this program block follows almost the same structure as the Basilisk propagation flow described in Section 5.1.3, but extends it with two additional model clusters: flight software and recorder flushing. The simulation models are also distributed across separate Basilisk processes rather than being scheduled on a single shared process. Environmental models are scheduled on `EnvironmentProcess`, each spacecraft is assigned its own `DynamicsProcess_i` and `FswProcess_i` for its dynamical spacecraft components and GNC system, and the data flushing model is scheduled on `RecorderFlusherProcess`. A comprehensive overview of the process-task-model hierarchical structure for this simulator is shown in Figure .0.3, which illustrates how the different models are distributed across all processes. Once initialization is complete, the Basilisk simulation is executed for the configured duration, and periodically write selected

simulation data to file to clear data buffers. The resulting total per-spacecraft output contains various time series and mission-level data used in the later lifetime analysis.

6.2 Autonomous Mission Operational Modes

The GNSS-R Formation Flight Simulator implements the mission-level operational constraints through a rule-based operational mode switching logic. This logic determines which high-level activity each spacecraft should perform at a given time, and thereby replaces a predefined operations timeline with autonomous decision-making based on the simulated spacecraft state and environment. This is important because the feasibility of GNSS-R observations, communication sessions, charging periods, and formation-maintenance maneuvers depends on time-varying conditions such as battery state, illumination, ground-station visibility, and maneuver requests. By evaluating these conditions during simulation execution, each spacecraft can adapt its operational mode to the current mission state, providing a more realistic representation of autonomous small-satellite operations than a fixed sequence of scheduled activities. The following subsections will present how each operational mode is defined, how each mode effect the instantaneous mission objective, and finally, the logic governing the transitions between operational modes.

6.2.1 Mode Definitions

The simulator uses eight operational modes, each representing a distinct mission-related spacecraft activity. A summary of the modes, their enabled power-consuming modules, and their associated pointing objectives is provided in Figure 6.2.1. In *coast* mode, the spacecraft remains in a nominal idle state while keeping the reaction wheels enabled to maintain a Sun-oriented attitude, even when the Earth occludes the Sun. The *charge* mode similarly prioritizes Sun-pointing, but is used specifically to increase the stored battery energy. In *capture* and *cooperative capture* mode, the payload is activated, and the spacecraft is oriented for GNSS-R observation. This entails pointing the GNSS-R receiver directly down towards the Earth’s surface (nadir-direction), and the GNSS receiver in the opposite direction (zenith-direction) toward the GNSS constellation. In *communication* mode, the communication system is enabled and the spacecraft points toward a visible ground station. The *burn transit* and *burn* modes support formation-maintenance maneuvers by orienting the spacecraft thrust axis toward the desired maneuver direction. The former enables propulsion-system heating, while the latter represents active thrusting. Finally, *emergency* mode disables all nonessential loads except the onboard computer and battery heater, and does not impose an active pointing objective. A more detailed definition of each power-consuming module is described in Section 6.3, while the pointing modes are described in Section 6.4.1.

Table 6.2.1: Overview of operational modes and corresponding active variable power modules and pointing objectives.

Operational mode	Additional variable power modules	Primary objective	Secondary objective
Coast	–	Sun-pointing	Minimum drag
Charge	–	Sun-pointing	Minimum drag
Capture	Payload	Data capture	Sun-pointing
Cooperative capture	Payload	Data capture	Sun-pointing
Comms	Communication system	Communication	Sun-pointing
Burn transit	Propulsion heating	Burn-pointing	Sun-pointing
Burn	Propulsion thrusting	Burn-pointing	Sun-pointing
Emergency	–	–	–

Reaction wheels are active in all nominal modes and are omitted from the module column for readability. The constant power modules: OBC and Battery pack heater are active for all modes

6.2.2 Mode Switching Logic

The mode switching logic is built around a set of logical parameters, here referred to as switching parameters, which determine whether transitions between operational modes are feasible. These parameters are divided into three groups depending on what part of the simulation state they depend on. The first group consists of environmental switching parameters, which are determined by spatial or illumination conditions. The parameter `canCap` indicates whether scientific data capture is feasible. In a full GNSS-R mission simulator, this would require a GNSS satellite to be within the GNSS receiver field of view and the corresponding specular reflection point to be within the GNSS-R receiver field of view. Since GNSS satellite constellations are not modeled in this simulator, data capture is assumed to be geometrically feasible whenever it is requested. The parameter `canCom` is true when at least one ground station has direct line of sight to the spacecraft, and its elevation angle is greater than 10° above the horizon. Finally, `canChar` indicates whether charging is possible, which requires the spacecraft to be fully or partially illuminated by the Sun.

The second group consists of battery switching parameters, which determine whether the remaining battery energy is sufficient to permit different operations. In this group, `capBat` and `comBat` indicate whether the battery level is high enough to allow science capture and communication, respectively, while `critBat` and `exitCriticalBat`, define entry into and recovery from critically low battery operation. The third group consists of operational state parameters, which depend on the internal mission state rather than directly on the external environment or battery level. These include `maxNoCom`, which becomes true if the time since the previous communication session exceeds a specified threshold, `burnRequested`, which indicates that a formation-maintenance maneuver is required and `burnAttitudeReached`, which indicates that the spacecraft is sufficiently aligned with the requested burn attitude. Finally, `capPossible` and `comPossible` are introduced to simplify switching logic visualization. The prior is a combination of `canCap` and `capBat`, and is true whenever both switching parameters are true. Similarly, the latter is true whenever its components, `canCom` and `comBat` are true.

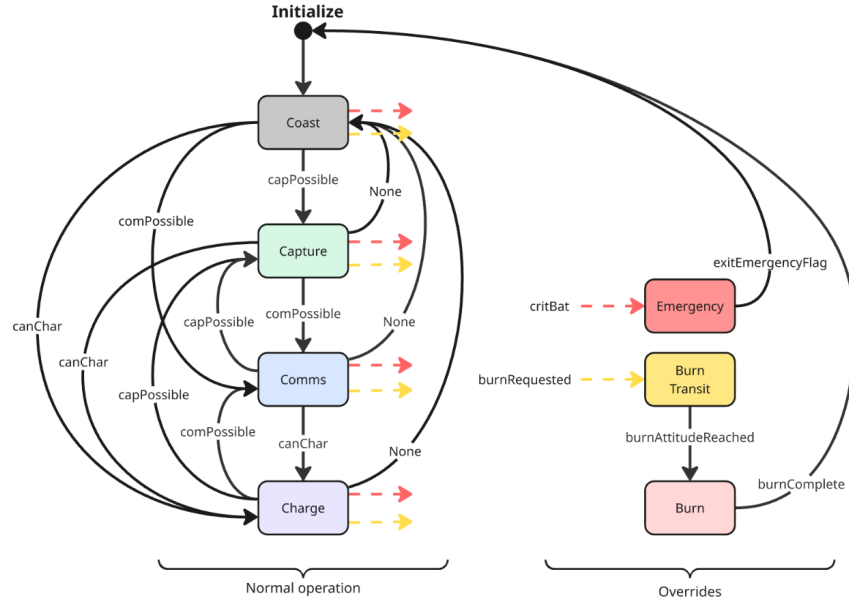


Figure 6.2.1: Mode switching logic flow diagram to visualize mode transitions. In cases where multiple feasible transitions from a mode exist, see the mode switching priority tables in Table 6.2.3 and 6.2.2 for normal operation and overrides, respectively

The resulting mode switching transition flow is illustrated in Figure 6.2.1. The left part of the diagram contains the nominal operational modes: coast, capture, communication and charge. These modes represent ordinary mission operation, where the spacecraft attempts to perform science observations, communicate with a ground station, charge the battery, or remain in a low-activity coast mode, depending on the currently feasible transitions. The labelled arrows indicate the switching parameters required for each transition. If none of the higher-priority nominal transitions are feasible, the spacecraft returns to or remains in coast mode. The right part of the diagram contains the override modes, which can interrupt the nominal operation logic. A critically low battery state triggers emergency mode, while a formation-maintenance request triggers burn transit, followed by burn once the required thrust-pointing attitude has been reached. After the emergency condition has cleared or the burn has completed, the logic returns to the nominal operation sequence.

Table 6.2.2: Override operation mode priority used when an override condition is active. The priority list is designed to resolve cases where multiple feasible mode transitions exist

Priority	Condition	Selected mode
1	<code>critBat</code> or (<code>Emergency</code> and not <code>exitEmergencyFlag</code>)	Emergency
2	<code>burnRequested</code> and not <code>burnAttitudeReached</code>	Burn transit
3	<code>burnRequested</code> and <code>burnAttitudeReached</code>	Burn
4	not <code>critBat</code> and not <code>burnRequested</code>	Use normal operation priority

Override conditions are evaluated before the normal operation mode priority logic. Emergency mode has the highest priority, followed by formation-maintenance burn modes.

Table 6.2.3: Normal operation mode priority used when neither emergency nor burn override is active. The priority list is designed to resolve cases where multiple feasible mode transitions exist

Priority	Condition when not <code>maxNoCom</code>	Selected mode	Condition when <code>maxNoCom</code>	Selected mode
1	<code>capPossible</code>	Capture	<code>comPossible</code>	Comms
2	<code>comPossible</code>	Comms	<code>capPossible</code>	Capture
3	<code>canChar</code>	Charge	<code>canChar</code>	Charge
Fallback	–	Coast	–	Coast

The selected mode is assigned by evaluating the listed conditions from highest to lowest priority. The priority between capture and communication depends on whether the maximum allowed time without communication has been exceeded.

Although Figure 6.2.1 shows the allowed transitions between operational modes and the switching parameters associated with each transition, it does not by itself define which mode is selected when several transitions are feasible at the same time. This ambiguity is resolved using the mode priority tables. If the battery has reached a critically low level or a formation-maintenance burn has been requested, the switching logic follows the override mode priority in Table 6.2.2. These override modes are evaluated before all nominal operations, ensuring that emergency battery protection and maneuver execution take precedence over capture, communication, charging and coasting. If neither emergency nor burn override conditions are active, the selected mode is instead determined by the normal operation mode priority in Table 6.2.3. In this case, the priority order depends on whether the maximum allowed time without communication has been exceeded, such that communication is prioritized over capture when `maxNoCom` is true.

6.3 Electronic Power System (EPS)

Power management is one of the key operational constraints for autonomous remote sensing satellite missions, since the available battery charge determines which operational modes can be safely executed. In the GNSS-R Formation Flight Simulator, the battery state therefore influences the spacecraft guidance logic by constraining transitions between internal operational modes, which is described in Section 6.2. This section, however, focuses on how electrical power generation, storage and consumption are modeled for each spacecraft in the formation. Since these quantities depend directly on the assumed spacecraft hardware, the EPS model assumes the spacecraft component configuration presented in Section 7.1.

6.3.1 EPS Model Overview

The EPS model is implemented to track the high-level electrical power flow for each spacecraft. Its purpose is to estimate the remaining battery energy during the mission, rather than to model detailed electrical bus dynamics. An overview of the per-spacecraft EPS structure is shown in Figure 6.3.1. Here, the green module represents power sources, red and yellow modules represent constant and variable power sinks, and the blue module represents battery energy storage. The power management module at the center represents the physical power distribution component included in the spacecraft configuration, but is not modeled with internal distribution logic, and does not contribute to the net power itself. The individual power generation, storage, and consumption models are described in the following subsections.

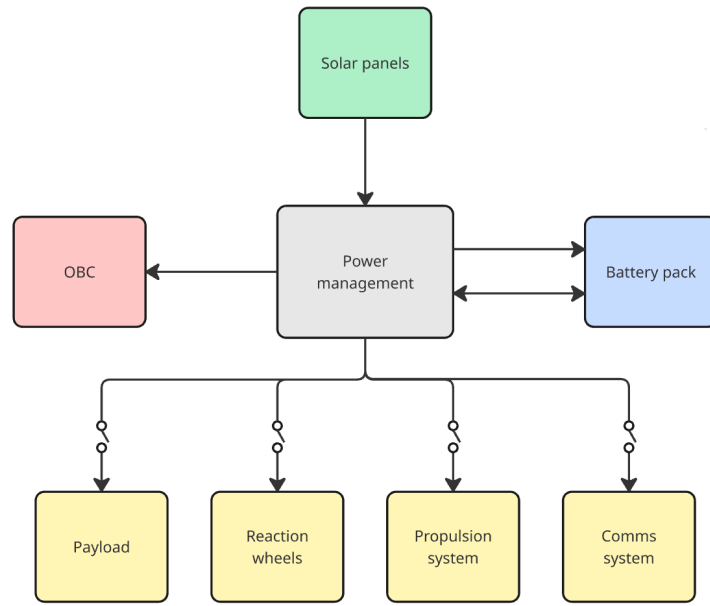


Figure 6.3.1: Overview of the per-spacecraft EPS

6.3.2 Power Generation & Storage

Electrical power generation is modeled using the Basilisk solar panel module, where each panel computes its generated power according to the simple solar panel formulation from Wertz and Larson in [26]:

$$W_{\text{sp}} = W_{\text{base}} C_{\text{eclipse}} C_{\text{panel}} (\hat{\mathbf{n}} \cdot \hat{\mathbf{s}}) A_{\text{panel}}, \quad (6.1)$$

Here, W_{sp} is the per-panel generated electrical power, W_{base} is the incident solar power per unit area, C_{eclipse} is the eclipse or shadow factor, C_{panel} is the solar panel conversion efficiency, A_{panel} is the panel area, $\hat{\mathbf{n}}$ is the panel normal vector, and $\hat{\mathbf{s}}$ is the Sun direction vector. The generated power is proportional to the cosine of the angle between $\hat{\mathbf{n}}$ and $\hat{\mathbf{s}}$ through their dot product, which makes the panel output attitude dependant. In implementation, panel output is set to zero when the Sun is behind the panel or when the spacecraft is in eclipse.

The generated power from all solar panel modules is accumulated by the battery model. The battery energy rate of change is given by the instantaneous balance between total generated and consumed power,

$$\dot{E}_{\text{storage}} = W_{\text{generated}} - W_{\text{consumed}} \quad (6.2)$$

where $W_{\text{generated}}$ is the total electrical power produced by solar panels, W_{consumed} is the total power drawn by spacecraft subsystems, and E_{storage} is the stored battery energy. The net power $\dot{E}_{\text{storage}}$ is integrated over time by the Basilisk battery model to estimate the battery energy level throughout the simulation.

6.3.3 Power Consumption

The OBC is modeled as a constant power sink. This represents the baseline electrical load required to keep the spacecraft operational, independent of attitude, illumination conditions, or operational mode. The constant power draw C_{OBC} is therefore always included in the total consumed power.

$$W_{\text{OBC}} = C_{\text{OBC}} \quad (6.3)$$

The battery heater is modeled as an averaged constant power sink. Since no battery temperature sensor or thermal model is implemented, the heater duty cycle is roughly approximated by assuming that the battery heater is active for one half of each orbital period. Under this assumption, the heating contribution to the total consumed power is modeled using the nominal active heater power, $C_{\text{bat,heating}}$

$$W_{\text{bat,heating}} = \frac{1}{2} C_{\text{bat,heating}}, \quad (6.4)$$

The payload power consumption is modeled as a mode-dependent power sink. When the spacecraft is performing data capture, the payload is assumed to draw the constant power C_{payload} . Outside this mode, the payload is treated as inactive and does not contribute to the power consumption.

$$W_{\text{payload}} = \begin{cases} C_{\text{payload}} & \text{during data capture} \\ 0 & \text{otherwise} \end{cases} \quad (6.5)$$

The reaction wheel power consumption is modeled using the Basilisk reaction wheel power module presented in [27]. The expression accounts for a constant baseline power draw W_{base} and an additional term associated with wheel actuation. The actuation-dependent part scales with the mechanical power required by the wheel W_{mech} , which depends on the commanded wheel torque and wheel speed, and is corrected by the assumed conversion efficiency η_{e2m} .

$$W_{\text{RW}} = \begin{cases} W_{\text{base}} + \frac{W_{\text{mech}}}{\eta_{e2m}}, & W_{\text{mech}} > 0 \\ W_{\text{base}}, & W_{\text{mech}} \leq 0 \end{cases} \quad (6.6)$$

The propulsion system power consumption is modeled using three discrete draw states. During propellant heating, the propulsion system draws a constant heater power to prepare the system for maneuver execution. During an active thrust maneuver, the power draw is set to the maneuver power level, representing the electrical load associated with firing the propulsion system. Outside heating and maneuver execution, the propulsion system remains in an idle state with a lower constant standby consumption.

$$W_{\text{prop}} = \begin{cases} C_{\text{prop,heating}} & \text{during propellant heating} \\ C_{\text{prop,thrusting}} & \text{during thrust} \\ C_{\text{prop,idle}} & \text{otherwise} \end{cases} \quad (6.7)$$

The communication system is modeled as a mode-dependent power sink. During communication mode, the transmitter and associated communication hardware are assumed to draw the constant power C_{comms} . When the spacecraft is not communicating with a ground station, this contribution is excluded from the total consumed power.

$$W_{\text{comms}} = \begin{cases} C_{\text{comms}} & \text{during communication} \\ 0 & \text{otherwise} \end{cases} \quad (6.8)$$

Finally, The total consumed power W_{consumed} used in Equation 6.2 is obtained by summing the active constant and mode-dependent power sinks at each simulation time. A comprehensive overview of when each power-consuming module is enabled is shown in Table 6.2.1.

6.4 Guidance Navigation & Control (GNC) System

The GNC system provides the flight software functionality required to translate operational mode decisions into spacecraft motion. In the simulator, this includes generating attitude references for the active operational mode, tracking these references using a reaction-wheel-based MRP feedback controller, and scheduling formation-maintenance maneuvers when the leader-follower geometry drifts from the desired configuration. The following subsections describe these three functions at a high level, focusing on their role in the closed-loop mission simulation rather than the detailed Basilisk message connections used to implement them.

6.4.1 Pointing Guidance System

The pointing guidance system maps the active operational mode to the desired attitude tracked by the attitude controller. This mapping is necessary because the operational modes in Table 6.2.1 impose different mission-critical pointing objectives. In Sun-pointing modes, the spacecraft is oriented to expose the largest possible solar panel area toward the Sun, increasing its power generation. In data-capture mode, the payload geometry is prioritized by pointing the GNSS-R receiver nadir, while the GNSS receiver is pointed

zenith, consistent with the payload constraint introduced in Section 4.6.3. In communication mode, the radio antenna is pointed toward the selected ground station when a direct line of sight is available. During burn transit and burn modes, the attitude reference is instead supplied by the formation-keeping controller described in Section 6.4.3, which aligns the body-fixed thrust direction with the desired maneuver direction. Finally, for modes in which minimum drag is included as a secondary objective, the spacecraft is oriented to minimize the cross-sectional area normal to the velocity direction.

When a mode defines both a primary and secondary pointing objective, the primary objective is enforced first. In the simulator implementation, this corresponds to aligning a body-fixed vector with a desired inertial direction, which still leaves one rotational degree of freedom about the primary pointing direction. This remaining freedom is used to maximize the secondary objective. For example, in communication mode, the antenna vector is aligned with the selected ground station direction, while the remaining rotation about the antenna axis is chosen to maximize solar-panel illumination, provided that the relevant panel normal is not parallel to the antenna direction. However, the secondary objective can generally not be satisfied exactly, because the primary pointing constraint restricts the available attitude space. Let $\hat{\mathbf{v}}_p^n$ denote the desired primary pointing direction, and let $\hat{\mathbf{v}}_s^n$ denote the direction associated with the secondary objective, both expressed in ECI. The component of $\hat{\mathbf{v}}_s^n$ that can be used while preserving the primary pointing direction is obtained by projecting it into the $\hat{\mathbf{v}}_p^n$ -normal plane,

$$\mathbf{v}_{s,\perp}^n = \hat{\mathbf{v}}_s^n - \frac{(\hat{\mathbf{v}}_s^n \cdot \hat{\mathbf{v}}_p^n)}{\|\hat{\mathbf{v}}_p^n\|^2} \hat{\mathbf{v}}_p^n \quad (6.9)$$

Using the normalized projected desired secondary pointing direction $\hat{\mathbf{v}}_{s,\perp}^n$, the transformation to the ECI frame from the desired frame $\mathcal{F}^R$, is then given by

$$\mathbf{R}_R^n = [\hat{\mathbf{v}}_p^n \quad \hat{\mathbf{v}}_{s,\perp}^n \quad (\hat{\mathbf{v}}_p^n \times \hat{\mathbf{v}}_{s,\perp}^n)] \quad (6.10)$$

which can then be converted into the corresponding desired MRP parameterization used by the attitude controller described in Section 6.4.2. The resulting attitude reference therefore maximizes the secondary objective within the constraint imposed by the primary objective. Note that the guidance system only calculates the desired attitude, not the corresponding reference angular velocity or angular acceleration. Consequently, time-varying attitude references are tracked purely through feedback by the attitude controller, which can increase reaction-wheel actuation and therefore reaction-wheel power consumption.

6.4.2 MRP Feedback Pointing Controller

Each spacecraft must be able to track the attitude reference generated by the pointing guidance system for the active operational mode. This is achieved using Basilisk's `mrpFeedback` module, which implements a nonlinear attitude feedback controller based on the MRP attitude and body frame angular-rate tracking error. The commanded body torque is computed using the control law presented in [28]

$$\begin{aligned} \mathbf{L}^b = & -K\boldsymbol{\sigma}_e - [P]\boldsymbol{\omega}_e - [P][K_I]\mathbf{z}_e - [I_{RW}](-\dot{\boldsymbol{\omega}}_{R/n} + [\tilde{\boldsymbol{\omega}}]\boldsymbol{\omega}_{R/n}) \\ & - \mathbf{L}_{ff} + ([\tilde{\boldsymbol{\omega}}_{R/n}] + [K_I\mathbf{z}]) ([I_{RW}]\boldsymbol{\omega}_{b/n} + [G_s]\mathbf{h}_s) \end{aligned} \quad (6.11)$$

In this expression, $\mathbf{L}^b$ is the requested control torque expressed in the spacecraft body frame, $\boldsymbol{\sigma}_e$ is the MRP attitude tracking error, and $\boldsymbol{\omega}_e$ is the body-frame angular-rate tracking error. The controller gains are given by the proportional gain K , the rate-feedback gain matrix $[P]$, and the integral gain matrix $[K_I]$, while $\mathbf{z}_e$ denotes the integral error state. The matrix $[I_{RW}]$ represents the spacecraft inertia including the reaction-wheel contribution, $\boldsymbol{\omega}_{R/n}$ is the angular velocity of the reference frame relative to the inertial frame, and $\boldsymbol{\omega}_{b/n}$ is the spacecraft body angular velocity relative to the inertial frame. The reaction-wheel geometry matrix $[G_s]$ maps the wheel angular momentum vector $\mathbf{h}_s$ into the body frame. Finally, $\mathbf{L}_{ff}$ represents feedforward compensation for known external torques, such as modeled disturbance torques.

In the simulator implementation, the integral feedback is disabled by setting $[K_I] = 0$. In addition, the pointing guidance system only defines the reference attitude, while the reference-frame angular velocity and angular acceleration are set to zero. This gives $\boldsymbol{\omega}_{R/n} = \mathbf{0}$ and $\dot{\boldsymbol{\omega}}_{R/n} = \mathbf{0}$, which removes the reference-motion feedforward terms from Equation 6.11. The implemented attitude controller can therefore be written as the simplified PD-type feedback law

$$\mathbf{L}^b = -K\boldsymbol{\sigma}_e - [P]\boldsymbol{\omega}_e - \mathbf{L}_{ff}, \quad (6.12)$$

where the remaining feedforward term is only present when known compensation torques are supplied. Under the assumptions stated in the Basilisk module documentation, the tracking control law can be shown to be asymptotically stable using the provided Lyapunov candidate function.

6.4.3 Formation-Keeping Controller

As described in Section 4.5, the absence of corrective maneuvers causes the relative geometry to drift away from the desired formation. Formation control is therefore required in the GNSS-R Formation Flight Simulator because the scientific mission concept depends on maintaining a prescribed leader-follower geometry over time. In this thesis, this is achieved using the Basilisk `spacecraftReconfig` module, which is an extended version of the spacecraft formation reconfiguration model described in [12] to account for the under-actuated control problem with only one static thrust vector. The control objective is formulated in terms of the orbital element difference between the two spacecraft, as introduced in Section 4.5.1. With $\boldsymbol{\alpha}_L$ and $\boldsymbol{\alpha}_{F_i}$ denoting the leader and follower i orbital element vectors, respectively, the formation-relative orbital element difference is written as

$$\Delta\boldsymbol{\alpha} = \boldsymbol{\alpha}_{F_i} - \boldsymbol{\alpha}_L = [\delta a \ \delta e \ \delta i \ \delta\Omega \ \delta\omega \ \delta M_0]^T. \quad (6.13)$$

Using this notation, the control objective can be written as

$$\Delta\boldsymbol{\alpha}(t) = \Delta\boldsymbol{\alpha}_R, \quad t \rightarrow \infty. \quad (6.14)$$

The `spacecraftReconfig` module achieves this objective by computing a sequence of impulsive maneuvers that drive the current orbital element difference toward the desired target difference $\Delta\boldsymbol{\alpha}_R$ within approximately one orbital period [29]. The module should therefore be interpreted as an orbital element-based maneuver scheduler rather than a continuous Cartesian relative-motion controller. As will be described in further detail

in Section 7.1, the simulated spacecraft is configured with a single body-fixed thruster. Therefore, the module also computes an attitude reference to align the body thrust direction with the desired impulse direction before the maneuver is executed. In the simulator, the resulting maneuver request is passed through the operational mode logic before the thruster is commanded, such that formation-control burns remain coupled to attitude pointing, fuel availability, propulsion-system state, and battery constraints.

The module also introduces several assumptions and limitations that constrain how the spacecraft orbits and components are configured, and influence how the results should be interpreted. First, the method is formulated using classical orbital elements, and is therefore not applicable to near-circular or near-equatorial orbits where these elements become poorly defined. Second, the reconfiguration strategy is based on impulsive maneuver planning, while the Basilisk implementation approximates each impulse using steady thrust over a finite time interval. This results in non-optimal orbital reconfigurations, especially for weaker thrust-to-weight ratio configurations. Thus, the maneuver fuel consumption should therefore be interpreted as a mission-level approximation of the formation-maintenance cost rather than an exact optimal maneuver solution.

6.5 Simulator inputs

SHARED SIMULATOR CONFIGURATION

To address the research objectives of this thesis, two distinct simulators were developed. The first simulator, referred to as the Comparative Propagation Simulator, is used to evaluate the long time translational prediction accuracy of the Basilisk numerical propagator by comparing its propagated spacecraft states against a Skyfield SGP4 baseline. The second simulator, referred to as the GNSS-R Formation Flight Simulator, extends the Basilisk-based simulation framework to model the autonomous operation of a dual-satellite GNSS-R formation mission subject to realistic operational constraints, with particular emphasis on formation-keeping fuel consumption and mission lifetime estimation.

Although the two simulators differ in purpose and implementation complexity, they share the same Basilisk spacecraft configuration, orbital environment models, and numerical integration setup. This chapter therefore presents the shared physical and numerical foundation in more detail. This common configuration is used to ensure that the propagation behavior assessed in the Comparative Propagation Simulator is directly transferable to the GNSS-R Formation Flight Simulator. However, it is important to note that the shared models presented in this chapter only apply to the Basilisk-based parts of the simulation framework. The Skyfield SGP4 propagation used in the comparative analysis is instead determined by the TLE data and the SGP4 model, and is therefore not configured using the same spacecraft or numerical integration method.

7.1 Shared Spacecraft Configuration

7.1.1 Spacecraft Platform

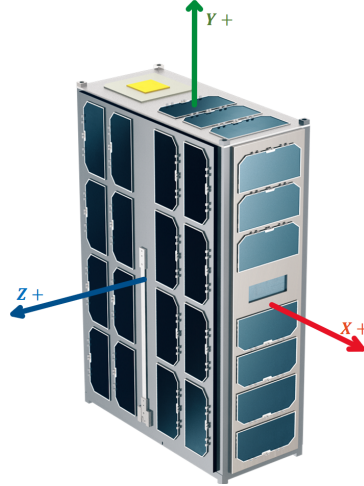


Figure 7.1.1: 6U spacecraft body frame definition used for both simulators. Adopted from [30]

7.1.2 Component Selection Method

7.1.3 Component Mounting Configuration

Table 7.1.1: Mounting configuration for spacecraft components.

Subsystem	Component	Mounting configuration
Structural	Frame	-
OBC	OBC	$\mathcal{O}_b$
EPS	Power management	$\mathcal{O}_b$
EPS	battery	$\mathcal{O}_b$
EPS	Solar panels BIG	Z+
EPS	Solar panels SMALL	X+ and X-
Comms	Radio Antenna	Y+
Payload	GNSS-R Receiver	Z-
Payload	GNSS Receiver	Z+
AOCS	Propulsion System	X-
AOCS	RWs	$\mathcal{O}_b$
Misc	Wires and harnesses	-

Table 7.1.2: Spacecraft mass budget. The full component parameter overview is shown in the Appendix Table .0.1

Subsystem	Component	Mass [g]	Source
Structural	Frame	665	GomSpace NanoStructure 6U
OBC	OBC	240	GomSpace NanoMind HP MK3
EPS	Power management	80	GomSpace NanoPower (dock, 1ACU, 1PCU)
EPS	battery	500	GomSpace Nanopower BPX 100 Wh
EPS	Solar panels BIG	392	Endurosat 6U Solar panel
EPS	Solar panels SMALL	2×147	Endurosat 3U Solar panel
Comms	Radio Antenna	2	Endurosat X-band patch antenna
Payload	GNSS-R Receiver	 	
Payload	GNSS Receiver	114	Endurosat GNSS patch antenna L1 + L2
AOCS	Propulsion System Dry	1430	VACCO ArgoMoon propulsion system
AOCS	RWs	4×144	Cubespace CW0162
Misc	Wires and mounting	100	
System mass:		4393	
20% margin:		879	
Total dry mass:		5272	
Propellant:		635	
Total wet mass:		5907	

7.1.4 Mass Budget & Inertia Estimate

Assuming uniform mass distribution, a perfect cuboid shape, and the body frame definition illustrated in Figure 7.1.1, the spacecraft moment of inertia matrix can be estimated to

$$J = \begin{bmatrix} 0.049221667 & 0 & 0 \\ 0 & 0.024610833 & 0 \\ 0 & 0 & 0.063988167 \end{bmatrix} \quad (7.1)$$

7.2 Shared Numerical Integration Configuration

The dynamics in both Basilisk-based simulators are propagated using the same numerical integration configuration. This choice is based on the numerical sensitivity analysis conducted in the specialization project [4], where several Basilisk integration settings were compared for the same orbital propagation scenario. The results showed that the low-step-size RK4 configurations and the variable-step RKF45 and RKF78 integrators converged toward the same Earth-relative and Formation-relative solutions. In contrast, fixed-step RK4 configurations with step sizes larger than 5 seconds produced more visible deviations. The study therefore concluded that numerical integration uncertainty was insignificant compared to the differences introduced by orbital perturbation selection over a 96-hour time span.

Although these results suggest that several of the tested numerical integration configura-

tions would be sufficient for the translational propagation problem alone, the simulators developed in this thesis impose stricter requirements on the numerical setup. In the Comparative Propagation Simulator, the inclusion of the NRLMSISE-00 atmosphere model introduces atmospheric density variations that are not present in the simpler propagation configurations considered previously. In the GNSS-R Formation Flight Simulator, the same orbital propagation is further coupled to fast attitude dynamics, actuator models, power generation, and autonomous mode switching. These additional models introduce faster dynamics, making it necessary to choose a conservative integration configuration that remains robust across both simulators.

Based on these considerations, both Basilisk-based simulators use the RKF78 variable-step integrator with a maximum step size of 0.5 seconds. A variable-step integrator is selected because it provides high accuracy during dynamically demanding parts of the simulation, such as attitude transitions and formation-keeping maneuvers, while allowing longer internal step sizes during less dynamic coasting or idle periods. This adaptivity is important for reducing computational cost over the long simulation horizons considered in this thesis. The RKF78 method is chosen because a high-precision integration method is desirable when propagating spacecraft trajectories over long time intervals, and because formation-control performance depends directly on accurate relative positioning between the leader and follower satellites. This justifies the increased computational cost per integration step. Finally, the maximum step size is limited to 0.5 seconds as a numerical safeguard against instability in the reaction wheel actuator dynamics used in the GNSS-R Formation Flight Simulator. Although this constraint is only required for the full formation-flight simulator, the same maximum step size is also used in the Comparative Propagation Simulator to ensure that results from the two simulators remain directly comparable.

VALIDATION & MISSION ANALYSIS METHODOLOGY

8.1 Propagation Validation Methodology

8.2 GNSS-R Mission Analysis Methodology

8.2.1 Mission Lifetime Definition

8.2.2 Evaluation Metrics

8.2.3 Monte Carlo & Scenarios

RESULTS

Plotting ideas for part 2:

- The big important plot: Fuel level over time
- It would be interesting to see natural formation drift, and calculate delta-V requirements to compensate for this drift to estimate lifetime. This estimate could be used to validate results.
- Show the relative differences in applied perturbation acceleration from differential J2 and drag to explain drift rate
- Each subsystem should be validated in addition to showing the full fuel consumption over simulation duration:
 - RW speed/torque step response
 - Thruster force output as a function of fuel level (/tank pressure)
 - MRP feedback controller tracking error and/or step response (would also be interesting to see a step response for different fuel levels)
 - Decomposed net power and battery level (should also show the illumination factor in the same/adjacent plot and the FSW state)

CHAPTER

TEN

DISCUSSION

IMPORTANT!

I have done some initial testing using the Basilisk Station Keeping example, and it seems like a strict CPO or Concentric formation is unachievable for the spacecraft reconfig module. If I have time, I should absolutely implement a HCW/ROE based controller for formation control. If I don't have time, I should just argue that this should be expanded in the future, and make a solid discussion about how my fuel consumption estimate is in comparison to the more realistic HCW case.

CHAPTER

ELEVEN

CONCLUSIONS

11.1 Conclusion

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APPENDICES

A - Github repository

All simulator code and LaTeX-files used in the thesis provided by the GitHub repositories linked below. Note that both the Comparative Propagation Simulator and GNSS-R Formation Flight Simulator are contained within the *Simulator Repository* link.

Simulator Repository

- <https://github.com/chystad/formation-flight-gnssr-simulator>

Master's Thesis LaTeX Repository

- <https://github.com/chystad/master-thesis-paper>

B - Simulator Flow Diagrams

Comparative Propagation Simulator

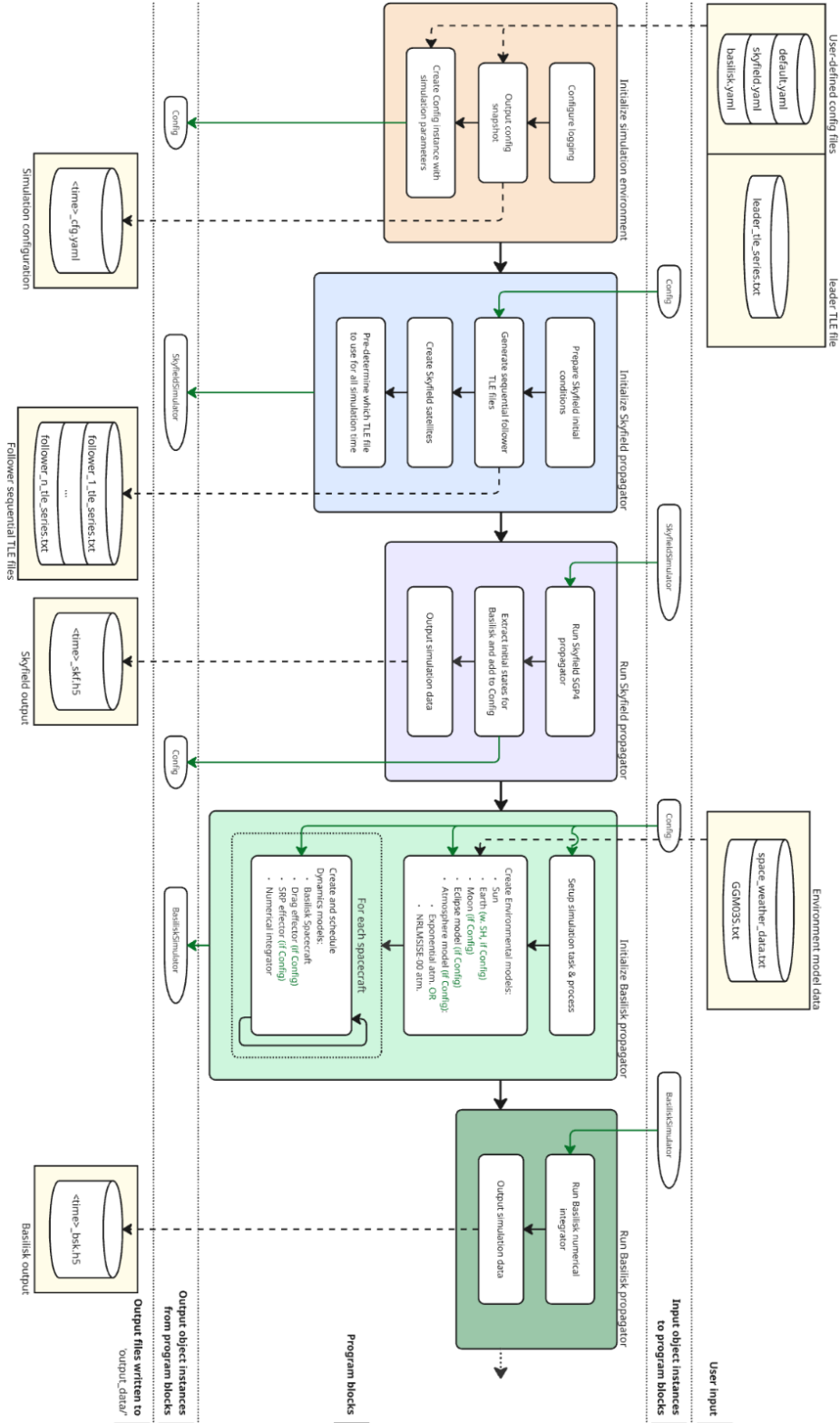


Figure .0.1: Full size detailed Comparative Propagation Simulator flow diagram

GNSS-R Formation Flight Simulator

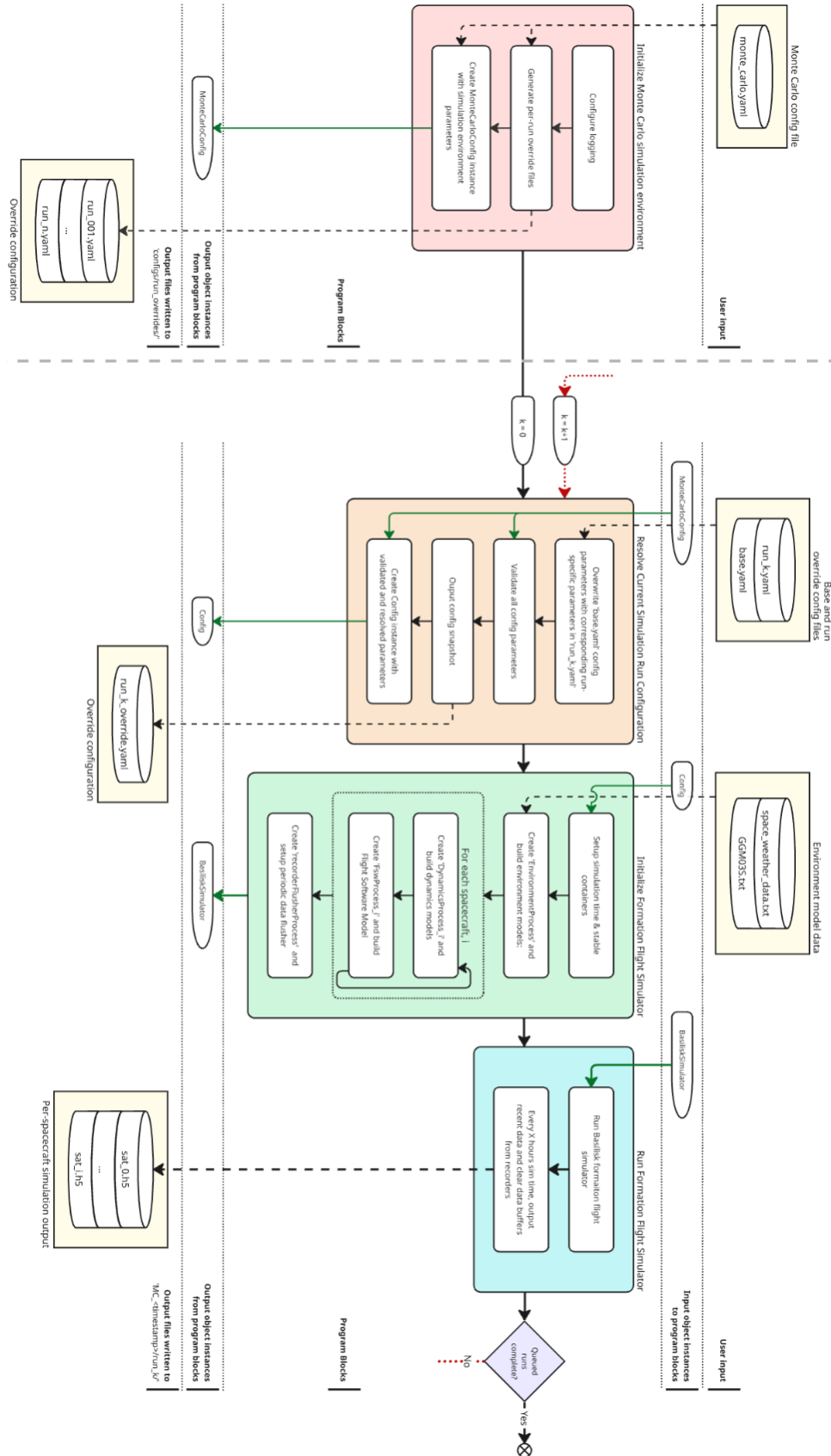


Figure .0.2: Full size detailed GNSS-R Formation Flight Simulator flow diagram

C - Spacecraft Component Parameters

Table .0.1: Overview of spacecraft components and relevant parameters.

Component	Mass [g]	Power consumption	Important parameters	Based on
Frame (Structural)	665	–	Dim. [mm]: $226.3 \times 340.5 \times 100.0$	GomSpace NanoStructure 6U
Power distribution (EPS)	80	–	- Dim. [mm]: $90 \times 96 \times 19.8$ - Only support one single connected battery	GomSpace NanoPower dock + 1ACU + 1PCU
Battery (EPS)	500	6 W (Heater)	- Dim. [mm]: $93 \times 86 \times 41$ 100 Wh, with 86 Wh in desired operating range 3500 mAh, with 3000 mAh in desired operating range	NanoPower BPX 100 Wh
Solar panels BIG (EPS)	392	19.2 W (Max)	- Dim. [mm]: 200×300 30% efficiency	EnduroSat 6U solar panel
Solar panels SMALL (EPS)	2×147	8.4 W (Max)	- Dim. [mm]: 100×300 30% efficiency	EnduroSat 3U solar panel
RWs (AOCS)	4×144	0.336 W (avg. at 2000 rpm) 7.2 W (Max)	- Dim. [mm]: $46 \times 46 \times 24$ 16.2 mNms at 6 krpm 10,000 rpm max 7 mNm max torque 4 in pyramid configuration	CubeSpace CW0162
Propulsion system (AOCS)	1430 dry 2065 wet	< 1 W (idle) 20 W (warmup) 4.3 W (operating)	- Dim. [mm]: $99.8 \times 92.8 \times 143.5$ - Hybrid Micro Propulsion System (MiPS) 1 100 mN green monopropellant thruster 425 mN cold gas thrusters 783 Ns total impulse ~ 133 Isp	VACCO ArgoMoon propulsion system
OBC	240	2 W (idle)	Dim. [mm]: $95 \times 95 \times 24.2$	GomSpace NanoMind HP MK3
Radio antenna (Comms)	2.2	4 W (max)	- Dim. [mm]: $24 \times 24 \times 5.6$ - Single X-band patch 74° half power beamwidth 6 dBi gain - Up to 4 W	EnduroSat X-band patch antenna

Continued on next page

Component	Mass [g]	Power consumption	Important parameters	Based on
Zenith GNSS antenna (Payload)	114	idk	• Dim. [mm]: $70 \times 70 \times 10$ • L1+L2 • 100–105 deg HPBW • 3.3–4.0 dBi gain • RHCP • Passive	GNSS patch antenna L1+L2
Nadir GNSS-R antenna (Payload)	idk	idk	• 12 dBi gain • 3×2 microstrip patch array • LHCP	–

D - Simulator Basilisk Process-Task-Model Structure

Comparative Propagation Simulator

GNSS-R Formation Flight Simulator

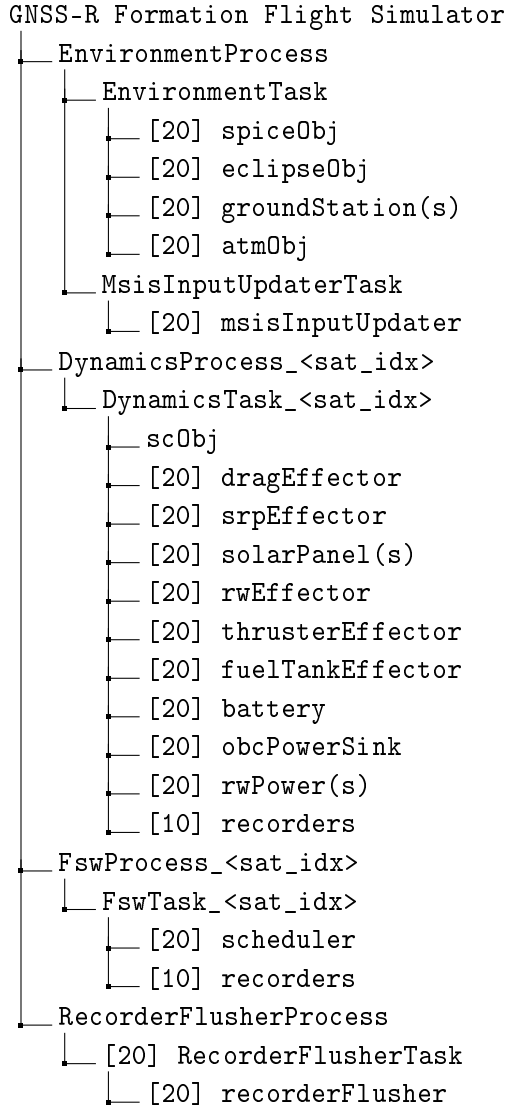


Figure .0.3: TODO: Update and verify after simulator update. Hierarchical process-task-model structure of the GNSS-R formation flight Basilisk simulator. The number in square brackets denote the model priority, where higher priority models are updated first. Same-priority tasks are orderly updated from the top-down within a task. “<sat_idx>” represents per-spacecraft processes and tasks.